

The University of the West Indies

Department of Computing & Information Technology

Source-Traceable Research Intelligence

Evidence-Grounded Discovery, Question Answering, and
Thesis-Extension Support for the TTLAB Corpus

MSc Data Science Project

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Editable manuscript sources, evidence manifests, and generated metrics are version controlled.

Abstract

Publication archives expose bibliographic records but do not directly support full-text discovery, evidence-linked question answering, author/topic exploration, or the formulation of bounded thesis extensions. This thesis presents the design and controlled offline evaluation of a source-traceable research-intelligence platform for a TTLAB laboratory corpus. Source-traceable means that paper, chunk, page, section, provider, timestamp, warning, and review-state identifiers persist through retrieval and generated-output workflows; it does not mean that every claim is factually correct or entailed.

The artefact-oriented methodology uses source-derived AI-reviewed silver evidence. Its frozen technical snapshot contains 134 catalog records, 96 eligible full-text papers, and 719 eligible chunks after excluding 2 PDF/title mismatches. Keyword, feature-hashing, and pinned learned-dense indexes cover the complete eligible set. The independent publication-and-rights gate is stricter: no paper in the audited snapshot was approved for public search, so anonymous Search, Ask, Finder, and Explorer fail closed rather than substitute technical records.

Immutable historical v1 evidence retains the negative diagnosis that traceable extractive answers can still omit requested points, choose imperfect citations, and fail to abstain. A prospectively frozen protocol then evaluated selective QA, paired Finder ranking, positive-only topics, and an OCR fixture. Over not run held-out cases, answerability precision was not run (not run) and exact-locator citation precision not run (not run). Finder Hit@3 changed by not run (not run) over not run profiles. Known-positive topic recall was not run (not run) over not run cases; OCR fixture character error was not run. These not available (v2 not run) technical observations are not human validation, entailment, public effectiveness, novelty, feasibility, or student benefit. The contribution is an evidence-calibrated platform and reproducible evaluation package, not a validated research advisor.

Keywords: research intelligence, information retrieval, retrieval-augmented generation, scholarly recommendation, responsible AI, reproducibility.

Statement of Scope and AI Assistance

This thesis reports software engineering, document processing, controlled offline experiments, and AI-assisted formative review. No human participants were recruited. TTLAB authorized the project and use of the laboratory corpus; that authorization is not represented as research-ethics approval or as permission to redistribute third-party PDFs. OpenAI Codex assisted materially with repository inspection, code changes, test preparation, AI-silver label creation and verification, analysis, and manuscript editing. The reviewer identity is retained as `codex-ai-review`; all resulting labels are described as AI-reviewed silver labels rather than human judgments. Jarod Esareesingh is the named thesis author. Any institution-specific authorship declaration, conflict statement, or submission attestation remains an external submission requirement and is not fabricated here.

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List of Abbreviations

API	Application Programming Interface
CI	Confidence Interval
DOI	Digital Object Identifier
FTS	Full-Text Search
LLM	Large Language Model
MRR	Mean Reciprocal Rank
MVP	Minimum Viable Product
nDCG	Normalized Discounted Cumulative Gain
OCR	Optical Character Recognition
PDF	Portable Document Format
QA	Question Answering
RAG	Retrieval-Augmented Generation
RQ	Research Question
TTLAB	Laboratory/project identifier used by the source archive and platform; no unverified acronym expansion is asserted
UI	User Interface

Introduction

Laboratory publication pages answer a bibliographic question—what has been published—but place the remaining analytical burden on the reader. A student must locate accessible full text, compare methods and limitations, identify a feasible extension, and avoid mistaking a generated proposal for an author-stated research gap. Researchers and public or industry users face related problems when locating work by topic or identifying a possible collaborator. The TTLAB precursor automated publication collection and display, lay summarisation, notifications, and a manual podcast workflow, and identified RAG querying as future work [1]. This project extends that direction from a publication archive into a full-text, source-traceable research-intelligence platform.

The wider scholarly-discovery landscape includes autonomous citation indexing in CiteSeer [2], expert and topic mining in ArnetMiner [3], literature graphs and large corpora such as Semantic Scholar and S2ORC [4, 5], and open bibliographic infrastructure such as OpenAlex [6]. These services demonstrate the value of linked scholarly metadata at scale. The present project addresses a narrower setting: a local laboratory corpus, full-text evidence with page and chunk locators, student-facing extension structures, public explanation, and explicit review state.

RAG can condition generation on retrieved documents [7], but a citation-bearing interface is not itself proof of retrieval completeness, entailment, recommendation feasibility, or user benefit. This distinction motivates the thesis’s evidence boundary. Software tests establish implemented behavior. Corpus manifests establish coverage. Source-reviewed cases establish bounded empirical quality. Human usefulness, author endorsement, supervisor fit, and institutional approval require evidence that was not collected in this study. A second boundary separates the technical corpus used for lawful local engineering and offline evaluation from the independently approved public projection. The latter is empty in the audited snapshot; public endpoints therefore return an explicit no-public-evidence state rather than leaking or silently substituting technical records.

1.1 Aim

The aim is to design, implement, and reproducibly evaluate a source-traceable research-intelligence platform that converts a bounded laboratory publication snapshot into searchable full text, citation-bearing answers, evidence-separated thesis-extension suggestions, and reviewable topic/author relations, while making failures and evidential limits visible.

1.2 Objectives

The objectives were to:

- O1. construct a deterministic, page-aware pipeline from catalog records and permitted local PDFs to eligible chunks and authoritative indexes;
- O2. implement keyword, lexical feature-hashing, learned dense, and hybrid retrieval over one frozen corpus;
- O3. preserve source identifiers and review state through search, RAG, recommendation, generated-artifact, and user-interface workflows;
- O4. evaluate retrieval, claim support, citations, answer completeness, recommendation structure, controlled topics, author identity, generated outputs, and performance using persisted raw evidence;
- O5. implement review, privacy, authentication, security, accessibility, and data-freshness controls appropriate to a demonstrable public-facing prototype; and
- O6. provide one-command reproduction and a versioned sanitized release that excludes restricted full text and secrets.

1.3 Contributions

The thesis makes five bounded contributions:

- C1. a layered research-intelligence artefact joining deterministic acquisition, page-aware chunking, four retrieval modes, answer and recommendation workflows, controlled topic/author discovery, and attributable review;
- C2. an authoritative index-manifest design that binds corpus identity, ordered chunk/-source hashes, model configuration, code commit, index hash, and completeness state, with atomic writes and fail-loud validation;
- C3. reproducible source-derived evaluation sets and raw outputs, with immutable histori-

cal v1 diagnosis separated from a prospectively frozen selective-QA, Finder, positive-only-topic, and OCR remediation protocol;

- C4. empirical evidence that simple keyword retrieval had the highest historical held-out point estimates in this corpus, without an equivalence claim for dense retrieval, together with prospective evidence about answerability, exact locator attribution, recommendation ranking, and known-positive topic coverage; and
- C5. an evidence-calibrated responsible-AI interface and sanitized release process that separates source facts, paper-stated future work, inference, suggestions, AI review, and external human validation.

No new retrieval algorithm, public-corpus effectiveness, human-validated recommendation benefit, researcher endorsement, production-scale capacity, or cross-institutional generalisation is claimed.

1.4 Thesis structure

Chapters 2 and 3 define the problem and research questions. Chapter 4 positions the work. Chapters 5 to 8 describe the executed study, system, corpus, and implementation. Chapters 9 and 10 define and report the controlled evaluations. Chapters 11 to 13 interpret the findings, state threats and governance boundaries, and answer RQ1–RQ7. The appendices provide the complete RQ traceability matrix, reproduction commands, schema/API summary, and executed AI-review instruments.

Problem Statement

The practical problem is a mismatch between a publication archive and the questions its users need answered. A student may ask which paper matches a set of interests and constraints, what the paper actually did, which extension is new system advice, and how the extension could be evaluated. A researcher may want an accurate technical or public summary. A public or industry user may seek publications related to a problem or authors whose indexed work provides evidence of a possible connection. These answers require evidence distributed across full papers, metadata, authorship relations, and generated interpretation.

Four risks make a naive archive chatbot unsuitable. First, metadata-only retrieval omits methods, results, limitations, and future work present only in the paper body. Second, PDF acquisition and extraction can silently introduce missing text, scanned pages, malformed metadata, or document-identity errors. Third, generation can merge retrieved facts with inferred gaps and novel suggestions. Fourth, a polished interface can conceal incomplete indexes, weak abstention, low answer coverage, unresolved identities, or unreviewed outputs. RAG supplies evidence to a generator [7]; it does not eliminate these risks.

The research problem is therefore:

How can a small, locally operated system transform a laboratory publication snapshot into a reproducible and inspectable research-intelligence workflow in which full-text evidence, generated interpretation, review state, evaluation state, and residual uncertainty remain distinguishable?

2.1 Stakeholders and failure consequences

The task/risk mapping in Table 2.1 determines which evidence and warnings each interface must expose.

Table 2.1: Stakeholders, tasks, and principal failure risks.

Stakeholder	Primary task	Principal risk
Student	find papers and scope a feasible extension	system suggestion presented as source fact, or feasibility presented as validated
Researcher	improve discoverability and explanation	inaccurate, incomplete, or decontextualised representation
Public/industry user	connect a problem to papers or people	publication-derived topic relation presented as endorsement or current availability
Reviewer/admin	correct metadata and generated outputs	unauthorised mutation, unattributed decisions, or AI review confused with approval
Evaluator	assess retrieval, grounding, and usefulness	coverage or structural traceability mistaken for quality

2.2 Scope

The implemented scope comprises a 134-record catalog; permitted local PDFs; deterministic and optional OCR-aware extraction; page-aware chunks; SQLite persistence; React/Vite presentation; FastAPI services; keyword, feature-hashing, learned dense, and hybrid retrieval; an offline extractive answerer and optional local Ollama adapter; evidence-only and full extension workflows; controlled topic/author discovery; generated summaries and text podcast scripts; AI and human review states; and offline evaluation. The technical corpus supports authorized local processing; an independent paper-level publication/rights/extraction gate determines the smaller public projection.

The study does not include unrestricted crawling, paywall bypass, hosted proprietary generation, participant recruitment, supervisor validation, researcher endorsement, audio synthesis, capacity/load testing, or production deployment. The optional OCR path is implemented and exercised by a deterministic raster-only fixture, but the corpus records no OCR-completed pages; fixture error is not corpus OCR accuracy. The platform name uses the laboratory identifier present in repository and archive evidence without asserting an unverified acronym expansion.

2.3 Operational definitions

Source-traceable means that an output retains persistent paper, chunk, page/section, provider/model/configuration, corpus/generation, timestamp, warning, and review-state fields. *Structurally grounded* means required citation fields and current retrieved-chunk membership pass deterministic checks. Neither term is synonymous with factual correctness

or entailment. *AI-reviewed silver* denotes labels created and verified in two passes by the same declared AI reviewer process; it is not human gold data or human inter-rater agreement. *Eligible technical corpus* denotes records with verified generation-linked PDF/text/chunk identity under frozen inclusion rules. *Public projection* denotes the independently approved subset whose review, publication, rights, access, extraction, and public-index generation states all permit anonymous delivery.

2.4 Success criteria

Engineering success requires deterministic processing, complete authoritative indexes for every eligible chunk, stable provenance fields, explicit degraded states, protected review mutations, reproducible builds, and a sanitized release. Empirical success is evaluated separately: retrieval quality relative to baselines, atomic claim support, citation correctness and completeness, answer-point coverage, abstention, recommendation proxy criteria, controlled-topic performance, section-label quality, review consistency, and bounded performance. Human usefulness remains outside the evidential scope.

Research Questions and Objectives

The research questions separate data integrity, retrieval, answer quality, recommendation, discovery/review, reproduction, and interface governance. The earlier five-question structure overloaded recommendation with topic/identity review and combined reproducibility with human-interface claims. Seven questions now give each construct a primary outcome and an explicit evidence boundary.

- RQ1** With what coverage, structural quality, and observable failure modes can the catalogue snapshot be transformed into an eligible page-aware corpus and complete authoritative indexes?
- RQ2** How do keyword, feature-hashing, learned dense, heuristic hybrid, and development-tuned hybrid retrieval compare on held-out source-derived questions, and which heuristic terms materially change rankings?
- RQ3** To what extent does the offline extractive answerer produce source-supported claims, correct claim–citation links, answer-point coverage, and appropriate abstention?
- RQ4** Relative to an evidence-only baseline, what does AI-assisted proxy review reveal about the Thesis Extension Finder’s relevance, source/suggestion separation, feasibility, scope, risk, and evaluation-plan quality?
- RQ5** What quality and governance limits remain in controlled topic assignment, publication-derived author links, identity handling, and persisted generated-output review?
- RQ6** To what extent can the bounded artefact and sanitized evidence package be reproduced and operationally characterized on documented local hardware, and what external format compatibility is demonstrated?
- RQ7** To what extent do the implemented public and protected interfaces preserve evi-

dence, uncertainty, privacy, authentication, correction, failure recovery, and automated accessibility contracts?

3.1 Research objectives

The questions correspond to seven objectives:

1. establish one authoritative corpus/index boundary that fails loudly on missing, mismatched, partial, or stale state;
2. compare retrieval families under one controlled split, with tuning and post-selection analysis clearly separated;
3. evaluate answer support, attribution, completeness, and selective response as distinct constructs;
4. implement and evaluate a thesis-extension workflow that keeps paper facts and system advice separate;
5. provide inspectable topic/author discovery and attributable generated-output correction without unsafe identity or approval claims;
6. bind code, corpus, model, experiment, document, and release evidence while characterizing local operational cost and external-format limits; and
7. trace user-facing requirements through APIs, interface states, protected mutations, and automated tests without treating software verification as human usability evidence.

3.2 Research-question design

Table 3.1 states the authoritative evidence and bounded conclusion type for each question. Its purpose is to prevent evidence migration: an inventory count cannot prove effectiveness, an AI-proxy review cannot prove student benefit, and a passing interface test cannot prove accessibility conformance.

Table 3.1: Research questions connected to executed method, evidence, and conclusion type.

RQ	Method	Authoritative evidence	Conclusion type
1	technical eligibility/i- dentity/generation audit; extraction/OCR diagnostics; public-projection and manifest validation	Phase 1 evidence; database integrity; active index generations; section cases; remediation OCR fixture	technical coverage/integrity, public-gate state, and observed data failures
2	historical v1 development/test evaluation; baselines; tuning; ablation; paired statistics	immutable raw per-query rankings; retrieval summary, statistics, and error taxonomy	bounded post-selection retrieval diagnosis and heuristic sensitivity
3	historical atomic diagnosis plus prospective source-first selective-response evaluation	v1 claim/link/point labels; prospective frozen cases, structural outputs, two-pass decisions, intervals	bounded extractive QA and repaired answerability/locator behavior, not entailment or human usefulness
4	historical proxy diagnosis plus prospective paired evidence-only/full comparison and sensitivity	v1 criteria; prospective frozen profiles, paired rankings, contract checks, profile-level intervals	proxy ranking/contract behavior, not novelty, feasibility, supervisor fit, or student benefit
5	historical topic comparison plus prospective positive-only recall; identity/source and stored-output review	historical/positive-only topic outputs; author audit; review events/idempotence	inspectable discovery/review behavior, not topic precision, expertise, public behavior, endorsement, or human approval

RQ	Method	Authoritative evidence	Conclusion type
6	repeated benchmark; full/quick reproduction; release scan; external open-corpus sanity check; document preflight	Phase 6 raw results; reproduction and release manifests; Europe PMC manifest; PDF checks	bounded local reproducibility, operational characterization, and format compatibility
7	requirement–API– view–test tracing; public privacy probe; role/authentication tests; failure-state and responsive/acces- sibility checks	route/API contracts; runtime probe; backend security tests; frontend com- ponent/E2E/accessibility tests	implemented evidence-interface governance, not user satisfaction, WCAG conformance, or production readiness

3.3 Claim boundaries

RQ1 concerns a generation-linked technical subset and cannot make unavailable or unapproved records publicly searchable. RQ2’s immutable historical v1 comparison does not establish broader-corpus effectiveness or equivalence. RQ3 evaluates deterministic extractive configurations; its intervals exclude AI-label uncertainty and cannot represent every RAG generator. RQ4 evaluates source-derived profiles and an AI proxy; novelty, data access, time, prerequisite mastery, and supervisor suitability remain external. RQ5 measures known-positive controlled labels, source-backed relations, and review mechanics; it does not establish topic precision, current author identity, availability, endorsement, public behavior, or human approval.

RQ6 reports concurrency-one local measurements, restricted-content-aware reproduction, and a three-article external XML check. It does not establish service capacity, PDF transfer, or broad generalization. RQ7 verifies coded contracts and automated regressions. It does not establish comprehension, trust, satisfaction, accessibility conformance, or usability with assistive technology. The complete method–evidence–result–conclusion mapping appears in Appendix A.

Literature Review

4.1 Design science as the study frame

Design-science research treats the construction and evaluation of an information-technology artefact as a means of producing knowledge about an identified problem. Hevner et al. distinguish this constructive paradigm from behavioral research and require an artefact, a relevant problem, rigorous evaluation, a research contribution, and effective communication [8]. Pef-fers et al. organize design-science research into problem identification, solution objectives, design and development, demonstration, evaluation, and communication [9]. These accounts are especially relevant to this thesis because the claimed contribution is not a new foundation model. It is a set of constructs, methods, software components, evidence contracts, and evaluation results assembled around a local research-discovery problem.

The distinction between building and evaluating is important. A route, database table, or source badge demonstrates implementation; it does not establish retrieval effectiveness, correct attribution, or user benefit. Conversely, a high offline metric does not demonstrate that a public interface communicates the metric's limits. This thesis therefore treats the repository and its executed evidence package as the artefact. Demonstration occurs through full-corpus workflows and route probes. Evaluation occurs at several units—corpus records, chunks, questions, claims, citations, profiles, topic labels, review events, and process stages—because a single end-to-end success score would hide where a failure arose.

The design-science frame also disciplines scope. The relevance cycle is the laboratory's need to expose a small, imperfect corpus to students and the public. The rigor cycle draws from scholarly retrieval, RAG evaluation, recommendation, human–AI interaction, and responsible-AI documentation. The design cycle repeatedly connects an observed failure to a design response and a bounded test. For example, stale indexes motivated hash-bound manifests; incomplete answers motivated answer-point review; unverified advice motivated evidence-only and generated modes; and unsafe correction motivated

actor-typed, append-only review events. Chapter 5 makes this mapping explicit rather than using “design science” as a retrospective label.

4.2 Scholarly discovery and research intelligence

CiteSeer demonstrated autonomous acquisition, citation extraction, and search over online papers [2]. ArnetMiner combined extraction and mining for expert and topic discovery [3]. Semantic Scholar’s literature graph, S2ORC, and OpenAlex illustrate complementary graph, full-text-corpus, and open-metadata infrastructures [4–6]. Their scale enables broad discovery and citation-network analysis, but it also differs from a laboratory collection whose source permissions, missing PDFs, incorrect attachments, and author variants may be insupportable one record at a time.

The TTLAB precursor automated publication collection, display, lay summaries, notifications, and manual podcast preparation, with RAG presented as future work [1]. The present artefact retains that public-communication motivation while changing the evidential unit from a publication-list entry or abstract to permitted full-text pages and chunks. It also makes exclusion visible: catalogue representation, local PDF availability, extraction success, eligibility, index coverage, and review state are separate fields. This separation prevents an absent document from disappearing silently and prevents a mismatched PDF from entering evaluation merely because a file exists.

A small system therefore has a different notion of completeness from a large index. It cannot claim global recall, but it can require that every eligible chunk appear in each authoritative representation and that every derived file bind to one ordered corpus snapshot. That form of internal completeness is an engineering claim, not a scholarly-coverage claim. The distinction motivates RQ1 and the manifest-validity checks used throughout the platform.

4.3 Information retrieval and evaluation

BM25 remains a strong lexical baseline [10]. Feature hashing projects sparse lexical features into a bounded-dimensional space [11]; because it has no learned language representation, calling it semantic retrieval would confuse implementation convenience with model capability. Sentence-BERT learns sentence embeddings suitable for cosine comparison [12]; Dense Passage Retrieval learns separate query and passage representations for open-domain QA [13]; and SPECTER incorporates citation structure into scholarly-document representations [14]. BEIR shows that zero-shot effectiveness varies materially across datasets and that dense retrieval should not be presumed superior [15].

These results support three methodological choices. First, lexical, hashed, learned-dense,

and hybrid modes are evaluated on identical cases and eligibility boundaries. Second, development tuning is separated from held-out testing. Third, effectiveness is reported with several ranking measures. Set Recall@ k answers whether all relevant papers were recovered, reciprocal rank emphasizes the first relevant result, and nDCG incorporates graded relevance and rank discount. None alone measures whether the corpus should have abstained.

Hybrid retrieval is not intrinsically better merely because it combines signals. Recommendation research similarly distinguishes hybridization strategies and their assumptions [16]. In this platform, the heuristic hybrid combines normalized keyword/vector scores with metadata, section, evidence, topic, and diversity terms; the tuned hybrid changes weights on development cases. Ablation is therefore necessary to determine whether a plausible term helps or harms. Paired tests and confidence intervals are used to resist reading small point-estimate differences as general superiority, while the absence of an equivalence test prevents the inverse claim that two modes are “the same.”

4.4 RAG, attribution, and answer completeness

RAG conditions generation on retrieved documents [7]. Fusion-in-Decoder aggregates evidence from multiple passages [17], while Self-RAG integrates retrieval and self-reflection decisions [18]. These architectures motivate evidence-aware generation, but a displayed citation may still be irrelevant to the nearby claim, and a source-supported sentence may fail to answer the question.

RAGAS and ARES propose automated dimensions and synthetic workflows for RAG evaluation [19, 20]. FActScore decomposes long-form text into atomic factual claims [21]. SummEval and G-Eval illustrate both the need for multidimensional review and the limitations of automatic or model-based judges [22, 23]. Following this literature, the thesis separates four constructs: whether a claim is supported somewhere in returned evidence; whether its attached citation is the correct evidence; whether expected answer points are covered; and whether the system declines questions outside the available evidence. Citation presence is recorded but is not allowed to stand in for the other three constructs.

Attribution also has an interface dimension. A paper ID without a passage requires the user to search the source again; a snippet without page or section context can be difficult to verify; and a “grounded” badge can imply more than a structural check established. The platform therefore returns paper, chunk, page, section, snippet, provider, model, warning, and review state. The later atomic review is intentionally stricter than the runtime structural verifier. This difference is not an inconsistency: the runtime can check locator integrity deterministically, while entailment and usefulness require a separate evidence process.

4.5 Abstention and selective response

Selective classification formalizes a reject option as a risk–coverage trade-off: a system may reduce error on answered cases by declining cases for which evidence is insufficient [24]. A RAG answerer is not identical to a classifier, so the theory is used here as a design analogy rather than a direct guarantee. The relevant principle is that answering every request is not automatically desirable when the corpus is bounded.

Abstention in research QA can fail at several levels. A query may be outside the catalogue; relevant papers may exist but lack eligible text; retrieval scores may be weak or contradictory; retrieved passages may not cover the requested answer points; or a provider may be unavailable. These conditions should not collapse into one generic error. The interface can return an explicit out-of-corpus notice, evidence-only results, a provider-fallback warning, or no generated answer. Evaluation must then report both coverage and error among answered cases, with enough unanswerable cases to estimate the trade-off. The present small unanswerable subsets are diagnostic, not sufficient for calibration.

4.6 Recommendation and human-centered advice

Research-paper recommendation combines content, citations, profiles, and hybrid signals; surveys identify reproducible evaluation and realistic user modeling as continuing challenges [25]. Publication history can represent recent research interests [26]. Broader recommender evaluation literature emphasizes that offline predictive measures, user tasks, and whole-system/user evaluation answer different questions [27]. A ranked paper may be relevant without producing a feasible project, and a well-formed idea may still be unsuitable for a particular programme or supervisor.

This distinction is central to the Thesis Extension Finder. Its profile contains interests, skills, available time, project type, data constraints, difficulty, preferred topics, and avoid-topics. Those inputs can support candidate retrieval and transparent heuristics. They cannot verify data permission, novelty, prerequisite mastery, institutional fit, or supervision capacity. The design therefore separates paper-supported baseline work; explicit paper-stated future work; an inferred or missing gap; and a system-generated suggestion. It also offers an evidence-only mode. The full output structures an MVP, stretch goals, risks, skills, data assumptions, and evaluation plan, but each remains advice requiring external confirmation.

Human-centered AI guidance recommends making capabilities and limits clear, supporting correction, showing contextually relevant information, and enabling user control [28]. Interactive machine-learning work likewise emphasizes user feedback throughout the system lifecycle [29]. Applied here, those principles favor profile echoing, editable inputs, visible

assumptions, evidence links, alternative evidence-only output, error recovery, and reviewer correction over a single opaque suitability score. The present study tests those contracts as software behavior; it does not substitute automated accessibility checks or AI-proxy ratings for a human usability study.

4.7 Topics, authors, and identity boundaries

Latent Dirichlet allocation represents documents as distributions over latent topics [30]. BERTopic combines transformer embeddings, clustering, and class-based term weighting [31]. Topic interpretability remains difficult to reduce to one numerical measure [32]. For a small curated corpus, the project retains an editable controlled vocabulary and evaluates it against a dense label-prototype alternative. The controlled method trades open-ended discovery for determinism and governance; precision, recall, coverage, and per-label support must determine how much that trade achieved.

Expert finding has been formalized through candidate- and document-centric language models [33,34]. In a public laboratory setting, however, an author–topic score may be misread as current expertise, availability, or willingness to supervise. The platform restricts author-topic evidence to eligible indexed publications, preserves aliases and unresolved candidates, and states that the relation is publication-derived. Identity merges require external confirmation because a false merge changes attribution across every dependent view.

4.8 Review, documentation, and responsible AI

Datasheets and model cards formalize provenance, intended use, evaluation, and limitation reporting [35,36]. Language-model risk taxonomies include misinformation and interaction harms [37]; the NIST AI Risk Management Framework organizes governance around mapping, measurement, management, and oversight [38]. These frameworks do not prescribe this application’s exact schema, but they motivate persistent source/configuration metadata, declared intended use, failure reporting, and accountable review.

The platform operationalizes those ideas through generated-content notices, actor-typed review states, authenticated mutations, append-only event hashes, transient public profiles, and a sanitized release. A correction is not a silent overwrite: the event identifies the target, prior and new state, actor role/type, rationale, time, and chain link. An AI reviewer can create an AI-reviewed state but cannot issue a human approval state. Content that cannot be reproduced safely remains marked for reprocessing rather than being deleted or relabelled as current.

4.9 Synthesis and research gap

Table 4.1 connects the literature-derived risk to the implemented response and the evidence still required. The table is a positioning device, not a capability-superiority claim.

Table 4.1: Literature-derived design tensions and this thesis’s bounded response.

Design tension	Implemented response	Evidence boundary
Corpus scale versus inspectable provenance	page-aware local corpus; eligibility states; hash-bound indexes	internal eligible-set completeness, not global scholarly coverage
Lexical versus dense/hybrid retrieval	common candidate/evaluation boundary; development tuning; ablation	small source-derived AI-silver test, not universal ranking superiority
Citation presence versus attribution	atomic support and claim–citation review; answer points	AI-reviewed silver evidence; no independent human reliability
Answer coverage versus abstention	distinct answerability and unanswerable outcomes; visible fallbacks	diagnostic unanswerable set, not calibrated selective risk
Paper relevance versus project suitability	evidence-only baseline; fact/gap/suggestion separation; explicit assumptions	synthetic profiles and AI-proxy review, not student/supervisor benefit
Topic/author discovery versus over-attribution	controlled vocabulary; per-label evidence; conservative identities	publication-derived association, not availability or endorsement
Generated output versus accountable correction	actor-typed states and append-only review events	implemented auditability, not completed institutional governance

Prior work supplies mature components for scholarly graphs, retrieval, recommendation, RAG, topic models, human–AI interaction, and documentation. The gap addressed here is their evidence-calibrated integration for a small laboratory: full-paper ingestion; paper/chunk/page/section traceability; source-fact, future-work, inference, and suggestion separation; evidence-only and generative recommendation modes; attributable review; and failure-oriented evaluation. The contribution is an engineered and evaluated integration, not a new foundational model, a complete bibliographic index, or a validated educational intervention.

Methodology

5.1 Study design

The study is an executed artefact-oriented design-science/engineering case study combined with controlled offline evaluation. The unit of engineering analysis is a versioned corpus-and-system snapshot. The unit of retrieval and QA analysis is a question case. The unit of recommendation analysis is a synthetic project profile. The unit of topic analysis is a paper case. Performance measurements use one operation or route workload per repetition. No human participants were recruited.

The study proceeded through an initial six-phase evaluation and a separate remediation evaluation. The initial phases covered corpus, extraction, identity and indexes; retrieval; atomic QA/citation review; recommendation, topics/authors and generated outputs; interface/security verification; and performance, release, and reproduction. Because that evidence was repeatedly inspected while implementation defects were being diagnosed, it is retained as immutable historical v1 evidence rather than re-labelled as prospective. The later protocol froze source-derived cases, splits, code, corpus, configuration, and output contracts before its held-out execution. Every empirical label produced by either process identifies `reviewer_type=ai`.

5.2 Design-science process mapping

The study follows the design-science requirements that an artefact address a relevant problem and be rigorously demonstrated and evaluated [8]. It also maps to the six activities in the Peffers et al. methodology [9]. Table 5.1 records how that mapping is evidenced in this repository. The phases are not claimed to have occurred as a perfectly linear waterfall; design and evaluation iterated when a test exposed a stale index, weak abstention, unsafe review boundary, or misleading presentation state.

Table 5.1: Design-science activities mapped to executed repository evidence.

Activity	Project interpretation	Executed evidence
Problem identification	publication lists do not support inspectable full-text discovery or bounded project formulation	problem statement; supervisor direction; corpus and interface audit
Objectives	preserve evidence, expose uncertainty, support discovery/advice, and remain demoable on a small lawful corpus	RQ/objective table; data and API contracts; scope exclusions
Design/development	build corpus/index, retrieval, Ask, Finder, explorer, evaluation, and review surfaces	versioned backend/frontend; schemas; manifests; code-native figures
Demonstration	execute primary workflows on the frozen authorized corpus	runtime probe; captured Ask/Evaluation states; Finder and Admin route traces
Evaluation	measure data integrity, retrieval, QA, recommendation, topics, review, performance, and interface contracts	phase artefacts; raw cases; statistics; backend/frontend tests; PDF preflight
Communication	report positive and negative findings with reproducible boundaries	thesis/paper; appendices; sanitized bundle; evidence and remediation registers

The central design claim is a pattern rather than an isolated screen: an output is reviewable when its source unit, transformation/configuration, uncertainty, and correction state survive every boundary that serializes or displays it. That principle produced several constructs: eligibility state for corpus membership; manifest state for indexes; citation and warning structures for answers; fact/future-work/gap/suggestion types for recommendations; controlled-label evidence for topics; canonical/alias states for authors; and actor-typed, hash-linked events for review. The evaluation asks whether each construct is populated and whether its implied quality claim survives stricter case-level analysis.

5.3 Requirement-to-failure logic

Implementation requirements were derived from observable failure modes rather than feature names. A “semantic search” requirement, for example, would prejudge the dense model; the operational requirement is instead to expose independently identified retrieval modes and compare them. A “grounded answer” requirement would conflate locators with entailment; the requirement is to return citations and warnings, then evaluate claim support, citation correctness, completeness, and abstention separately. A “thesis recommender” requirement could imply assignment authority; the requirement is to rank source evidence and place generated extensions behind a clearly labelled, optional boundary. The resulting falsification-oriented mapping is summarized in Table 5.2.

5.4 Evidence hierarchy and reproducibility controls

Claims were admitted only when supported by an authoritative source:

1. immutable or hash-linked manifests establish corpus, model, index, and release identity;
2. raw case-level outputs establish executed experimental observations;
3. evaluation code and unit tests establish metric definitions and implementation behavior;
4. database integrity checks and protected runtime probes establish current operational state;
5. automated frontend and security tests establish the tested interface contracts; and
6. compiled/rendered documents establish publication artefact quality.

Generated manuscript macros and tables are produced from frozen JSON artefacts by thesis/scripts/generate_manuscript_assets.py; their input and output hashes are recorded in thesis/

Table 5.2: Observed risk, design response, and falsifying evidence.

Observed risk	Design response	Evidence that can falsify success
wrong or unavailable document enters search	title/content eligibility and persistent unavailable/mismatch states	audit counts; excluded IDs; source hashes; extraction diagnostics
derived index silently covers a subset	ordered corpus/configuration manifest and startup rejection	coverage/hash mismatch tests; readiness state
retrieval label implies superiority	explicit mode names, shared protocol, tuning split, ablation	held-out rankings; paired statistics; negative comparisons
citation badge implies a correct answer	structured citations plus separate claim, link, point, and abstention review	partial/incorrect links; missing points; false-positive answers
paper facts become generated “future work”	typed baseline, explicit/inferred/missing gap, and suggestion fields	source-locator checks; boundary criterion decisions
topic/author link implies verified expertise	controlled evidence and publication-derived qualifier; conservative identity candidates	per-label errors; identity/source audit
correction erases history or AI impersonates a human	append-only actor-typed events and role-restricted states	mutation/role tests; event-chain and idempotence checks
passing UI appears equivalent to human validation	requirement/API/view/test trace plus explicit human-study boundary	automated failures; missing user/assistive-technology study

generated/manuscript_metrics_manifest.json. The remediation package is not completed or validated and its source revision prefix is not run. The generator refuses a partial package and imports every remediation numerical result from its validated macro file; hand-entered copies are rejected by manuscript validation. Tests and build counts are reported only as engineering verification, never as quality metrics.

The evidence hierarchy also separates committed application state from generated manuscript state. The manuscript manifest records the committed base commit/tree, the exact generator hash, each input hash, and each output hash. Generated files and compiled PDFs form an explicit dirty-output boundary; they are not silently attributed to the earlier performance execution revision. Corpus snapshot, model revision/hash, evaluation artefact revisions, manuscript source state, and PDF hash are related identifiers, not interchangeable meanings of “version.”

The immutable historical v1 artefacts retain their negative findings but are classified as post-selection diagnosis. The prospective remediation protocol instead froze source locators, answer points, absence probes, profile needs, known-positive topics, split IDs, code, corpus, configuration, and environment before held-out execution. It reconciles PDF-extraction-chunk generations on a temporary database, rebuilds keyword search, runs once, performs a shuffled second AI-silver review, retains disagreements, and recomputes cluster-bootstrap intervals. Same-process repeatability is not independent reliability or human validation; intervals omit label and corpus-selection uncertainty. Rights-sensitive answers, snippets, and recommendations remain ignored locally while structural outputs and hashes are versioned.

The run binds source not run to technical corpus not run (not run papers; not run chunks), scope not evaluated, and public-projection execution no. Technical eligibility requires matching extraction/chunk generations; public delivery additionally requires human approval, published state, cleared rights, searchable access, approved extraction, and a current public generation. Selective QA fixes keyword retrieval and the not run/not run path. Finder pairs evidence-only and not run rankings. not run is evaluated only for known-positive recall, and the not run fixture exercises the parser without claiming scanned-corpus accuracy.

5.5 Corpus freeze and inclusion rules

Catalog records were retained even when full text was unavailable. A paper entered the experimental full-text corpus only when its PDF was present, extractable, and its identity passed title/document checks. Suspected PDF/title mismatches were excluded, not repaired by assumption. Every eligible chunk’s identifier and source hash contributes to the ordered corpus snapshot hash. The authoritative corpus is corpus-04a010207327069a, comprising 96 papers and 719 chunks. The catalog additionally contains 36 records requiring text

acquisition/review and 2 mismatch exclusions.

5.6 Historical v1 AI-reviewed silver annotation

The historical retrieval cases were constructed from source passages and stratified by exact fact, methods, results, limitations/future work, title/author/venue, topic/application, comparison, broad intent, multi-paper synthesis, hard negative, and out-of-corpus intent. A creation pass recorded relevant paper IDs, supporting chunks/pages, rationale, and answerability. A later shuffled verification pass checked labels without access to system rankings. Creation/verification disagreements and adjudication were retained. Because the same AI reviewer process performed both passes, consistency is described as annotation consistency, not inter-rater reliability.

Historical QA review used a related 50-case set with 46 answerable and 4 unanswerable questions. Answers were segmented into atomic checkable claims; each claim–citation relation was assessed as supported, partial, or unsupported. AI-reviewed silver reference answer points were assessed separately. Recommendation and topic reviews used the same two-pass AI-silver principle. Human validation remains a limitation throughout.

5.7 Retrieval metrics

For a query q with relevant-paper set R_q and returned top- k paper set $S_{q,k}$, set recall and precision are

$$\text{Recall}@k(q) = \frac{|R_q \cap S_{q,k}|}{|R_q|}, \quad (5.1)$$

$$\text{Precision}@k(q) = \frac{|R_q \cap S_{q,k}|}{k}. \quad (5.2)$$

Hit@ k is one when $R_q \cap S_{q,k} \neq \emptyset$ and zero otherwise. Reciprocal rank is $1/r_q$ for the rank r_q of the first relevant paper, and zero when none is retrieved. nDCG@10 uses graded source labels and logarithmic rank discount. Unanswerable false-positive rate is the proportion of out-of-corpus cases for which a mode returns a positive result instead of abstaining. This corrects the earlier ambiguity in which a binary hit measure had been called recall.

5.8 Tuning, uncertainty, and statistical analysis

30 of 50 retrieval cases formed the development partition and 20 formed the held-out test partition. Keyword, feature-hashing, learned dense, and heuristic hybrid modes used identical

corpus, filters, candidate-pool depth, unique-paper cut-offs, and cases. Hybrid weights were selected on development MRR from a recorded search space; test labels were not used for selection. Individual and cumulative ablations disabled query expansion, metadata, section, evidence, topic, and diversity terms. Weight sensitivity varied keyword/dense proportions around the selected values.

Percentile bootstrap 95% CIs used 10,000 query-level resamples with fixed seeds. Paired randomisation tests compared the tuned hybrid with baselines on answerable test cases. Holm correction controlled multiplicity within metric and experiment families. These intervals reflect query-sample uncertainty, not uncertainty in AI-silver labels or corpus selection.

Null results are retained as results. A confidence interval spanning zero does not prove equivalence, and an adjusted p -value above the threshold does not prove no difference. Likewise, an ablation inspected after candidate selection is descriptive and cannot be used to retune on the held-out set. The reporting rule is therefore: identify the selection boundary, report point estimate and uncertainty, state whether the planned comparison rejected its null, and avoid converting failure to reject into equality or non-inferiority.

AI-label uncertainty is handled by disclosure and case retention rather than a fabricated numeric adjustment. The same declared AI process performed source-first creation and shuffled verification, so exact agreement measures repeatability under that procedure, not independent reliability. Disagreements, rationales, and source locators remain available for later human adjudication. Bootstrap intervals resample cases/profiles but hold labels fixed; every interpretation therefore states that correlated label error and corpus selection are outside the interval.

5.9 QA, recommendation, and topic metrics

Strict supported-claim rate counts only fully supported checkable claims. Citation correctness divides correct claim–citation links by all reviewed links; citation completeness measures citation presence, not correctness. Answer-point coverage divides fully covered points by all expected points. Abstention accuracy distinguishes answering answerable cases from declining unanswerable ones.

The historical recommendation experiment compared three evidence-only paper/passages with three full-finder recommendations for each of 28 profiles. AI-proxy criteria included relevance, source fidelity, fact/future-work/gap/suggestion separation, novelty caution, feasibility, MVP and stretch scope, risk, skills, evaluation plan, and proxy usefulness. Its paired contrast resampled profiles. Historical topic evaluation reported multi-label micro precision, recall, F1, exact match, and coverage for controlled lexical and learned-dense prototypes.

The later positive-only topic protocol deliberately narrows its construct to known-positive recall and case coverage; predictions beyond the listed positives are not counted as false positives.

5.10 Performance and external sanity methodology

The Phase 6 harness runs process-cold and warm repetitions at concurrency one for discovery, import, extraction, chunking, keyword/hashing/dense indexing, four retrieval modes, offline answering, recommendation, read-only APIs, frontend production build, and primary route loads. It reports median, linearly interpolated p95, failures, processed units, and maximum resident set size where available. “Cold” means a fresh process, not a flushed operating-system cache. The external sanity check reacquires three pinned CC BY articles from the official Europe PMC API, validates embedded license evidence, chunks them, and tests three fixed lexical-discrimination queries. This establishes only format compatibility and trivial discrimination, not external TTLAB effectiveness.

5.11 Ethics and authorization protocol

TTLAB authorized the project and use of the laboratory corpus. The study does not convert that authorization into a claim of ethics-board approval, blanket copyright ownership, or re-distribution rights. No participants were recruited. Student-profile fields are transient by default. Restricted PDFs, full extracted text, interface screenshots, private prompts, databases, indexes, and secrets are excluded from the sanitized bundle. AI assistance and reviewer identity are disclosed in the front matter and evidence manifests.

System Architecture and Design

6.1 Layered architecture

The four layers (Fig. 6.1) are a React route UI; FastAPI public-projection/protected APIs and security; ingestion, retrieval, selective QA, recommendation, discovery, review and evaluation logic; and SQLite plus immutable local generations. Only protected/local evaluation may select the technical corpus explicitly.

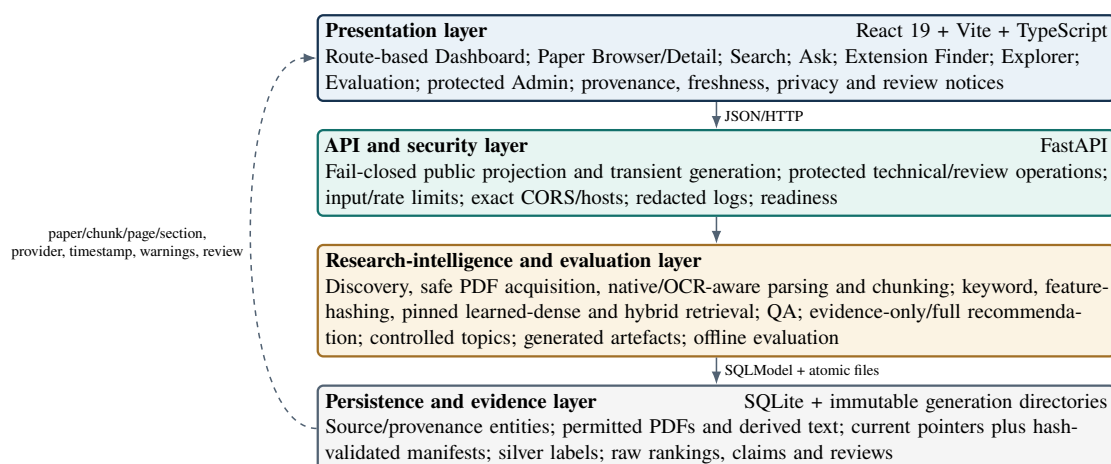


Figure 6.1: Layered system architecture. The dashed path represents provenance and review data returned from storage to public evidence displays. Text labels provide the non-color encoding.

6.2 Design principles

Six principles govern the design. First, derived index state is authoritative only when its manifest and active generation pointer match the current corpus. Second, technical eligibility never implies publication or redistribution approval. Third, source facts and generated interpretation use distinct fields and labels. Fourth, AI review has a distinct `ai_reviewed` state and cannot create human approval. Fifth, public requests are privacy-minimised and ad-

ministrative mutations are authenticated. Sixth, missing dependencies or evidence produce explicit unavailable, stale, partial, support_unverified, or needs-reprocess states instead of silent success.

6.3 Persistence and traceability model

Figure 6.2 summarises the entities. Papers/chunks store extraction, chunk and approved-public generations; outputs retain structured source identifiers. Controlled topics link papers and canonical identities. Each transition appends an actor-typed, request-attributed, hash-linked ReviewEvent, preserving history without treating AI review as human approval.

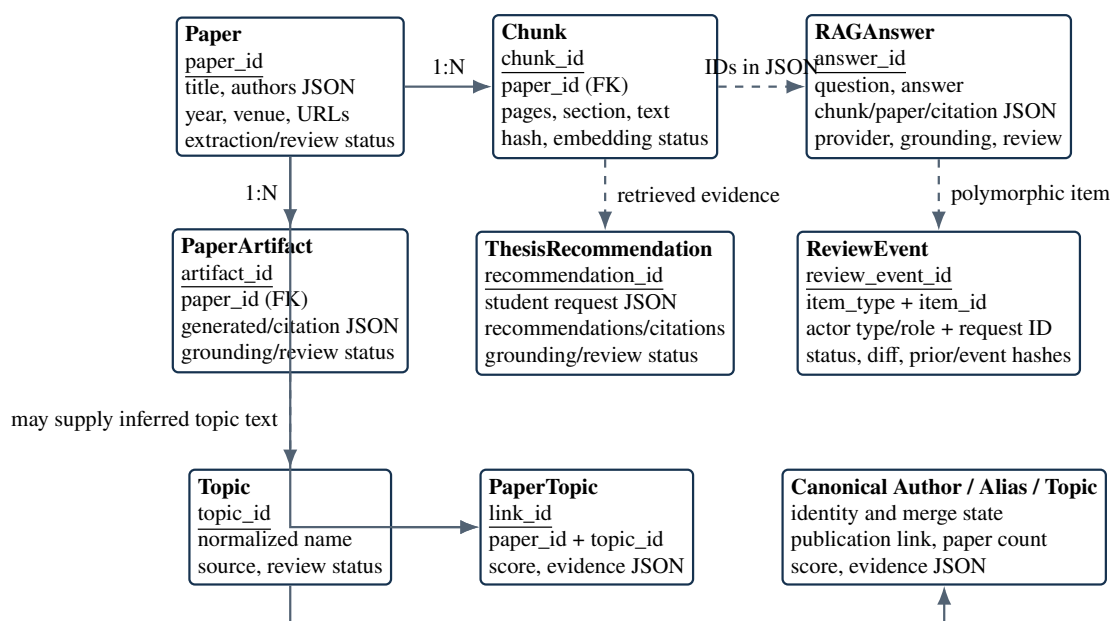


Figure 6.2: Principal persistence and provenance entities. Solid arrows denote ownership/foreign-key relations; dashed arrows denote evidence or polymorphic references.

6.4 Authoritative index design

Each vector manifest binds the corpus and ordered chunks/hashes to provider/model/revision, dimension/normalisation, configuration/code/index hashes, role, and completeness. A rebuild validates an immutable generation, commits matching database status, then atomically moves the current pointer; failure leaves the prior pointer active. Demo indexes cannot target the canonical location, and validators reject missing, partial, stale, or mismatched state.

Feature hashing is a 256-dimensional signed lexical projection, not learned semantics. Dense retrieval pins all-MiniLM-L6-v2 revision 826711e...a1def9, its artifact hash, 384 dimensions, normalization, CPU/window pooling, and license. Optional Ollama tags

are allowlisted by digest and checked before/after generation; absent response attestation is recorded as a mutable-tag limitation.

6.5 Security and privacy boundaries

Public reads/generation are separate from technical evaluation and protected mutations. Public metadata requires human approval, published state and cleared rights; searchable text also requires approved extraction/current public generation. No audited record met the full predicate, so anonymous evidence routes fail closed. Environment-digested bearer credentials, production HTTPS/host/CORS checks, role restrictions, size/rate limits, redaction, SSRF/redirect controls, and PDF limits address principal threats.

Student interests, skills, time, constraints, and topic preferences remain client-side/in-request and are not persisted by default. Public generated-output serializers omit private reviewer notes and draft corrections. This minimisation reduces privacy risk but does not replace an institutional privacy review for a deployed service.

Data Collection and Processing

7.1 Catalog and acquisition

The catalog contains 134 publication records derived from the TTLAB archive and stored in deterministic seed JSON. The discovery/import path normalises titles and URLs, assigns stable identifiers, preserves absent metadata as missing, and records provenance/review state. PDF acquisition uses allowed hosts and public IP resolution, bounded redirects, timeouts, content-length and byte limits, content-type and PDF-signature validation, and explicit failure categories. It does not bypass authentication, paywalls, or access controls.

The frozen audit records 98 available PDFs, 34 records with no PDF URL, 1 network error, and 1 not-found result. Of the available documents, 96 matched their catalog identity and entered the experimental corpus; 2 possible PDF/title mismatches were excluded. The remaining 36 catalog records are visible as needs-review/no-text records rather than being silently absent. Figure 7.1 shows the full pipeline, and Fig. 7.2 shows the resulting paper-level eligibility categories.

7.2 Extraction and OCR-aware diagnostics

PyMuPDF records page-aware native extraction and hashes; image-heavy pages may use an optional bounded OCR worker with explicit failure/provenance. Corpus OCR status is `not_requested`. A deterministic fixture produced character error not run through the same path, which is regression evidence rather than scanned-corpus accuracy.

The 98 extracted PDFs contain 698 pages and 455,601 native-extraction words. These totals describe the local source snapshot; they are not included in the sanitized release. The suspected identity mismatches account for 16 excluded chunks. The raw chunk table contains 735 rows, while the eligible experiment and authoritative indexes contain 719 rows.

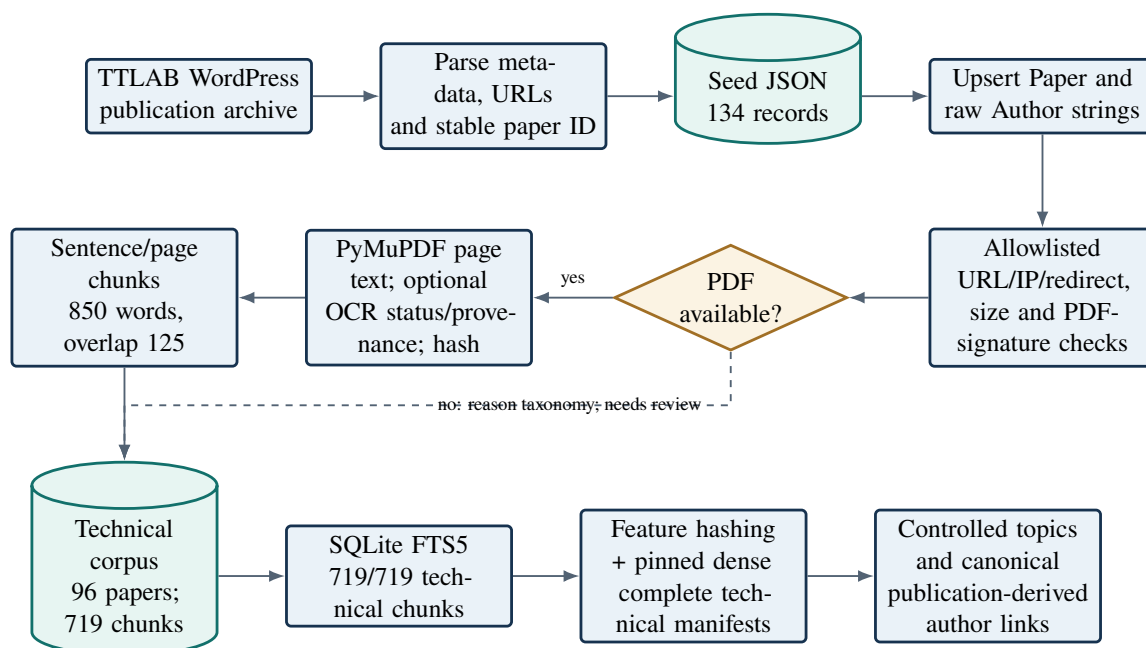


Figure 7.1: Deterministic ingestion and authoritative indexing pipeline. The optional OCR stage records status/provenance but was not used in the frozen corpus.

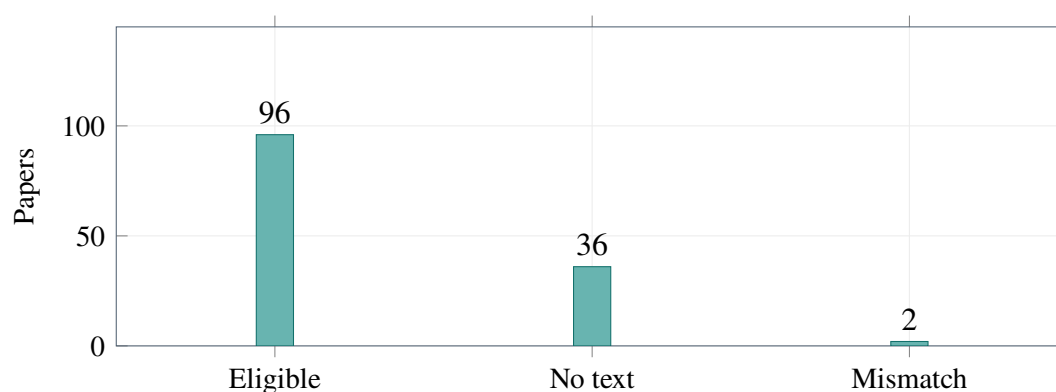


Figure 7.2: Frozen catalog eligibility: 96 eligible full-text papers, 36 records without eligible text, and 2 PDF/title mismatch exclusions. These are inventory counts, not quality scores.

7.3 Page-aware chunking and sections

Chunks retain pages and target 850 words, 5,000 characters, and 125-word overlap. Extraction manifests bind PDF/configuration/exact text bytes; chunk manifests bind that generation to serialized and database chunks. Reconciliation quarantines any disagreement and clears public approval instead of inventing provenance. Inferred sections preserve Unknown; 199 eligible chunks (27.7%) have that state.

A 40-case stratified AI-reviewed silver sample compared current section labels with source passages. Exact accuracy was 0.900, labeled coverage 0.900, macro precision 0.892, and macro recall 0.904. Seven creation decisions required revision. These results support using section labels as a reranking feature with caution, not as authoritative document structure.

7.4 Metadata and author identity

The malformed Click to View author parser artifact was removed by an idempotent migration. The schema now stores canonical author rows, aliases, identity/merge state, and review status. The audit found 136 author rows: 133 unresolved active canonical identities, two merged rows, and one invalid row. Exact alias collisions, eligible authorship mismatches, excluded-paper leakage, and prohibited expertise overclaims were all zero in the executed audit. 13 possible initial/full-name or spelling pairs remain deliberately unmerged pending authoritative confirmation.

DOI, abstract, keyword, venue, and date fields carry provenance and review fields. Unverified metadata remain empty. Author-topic links are derived only from eligible indexed publication authorship and paper topics; their UI wording explicitly disclaims availability, endorsement, supervision, or expertise beyond this corpus.

7.5 Corpus distribution and release boundary

The distribution in Fig. 7.3 characterises the seed, not eligibility. The sanitized release excludes PDFs, derived text, databases/indexes, raw answers/snippets/recommendations, and other restricted content because project authorization is not blanket redistribution permission. It retains safe metadata, schemas, structural results, labels, hashes, and instructions. Paper-level rights/review/generation approval is independent; the current public projection is empty.

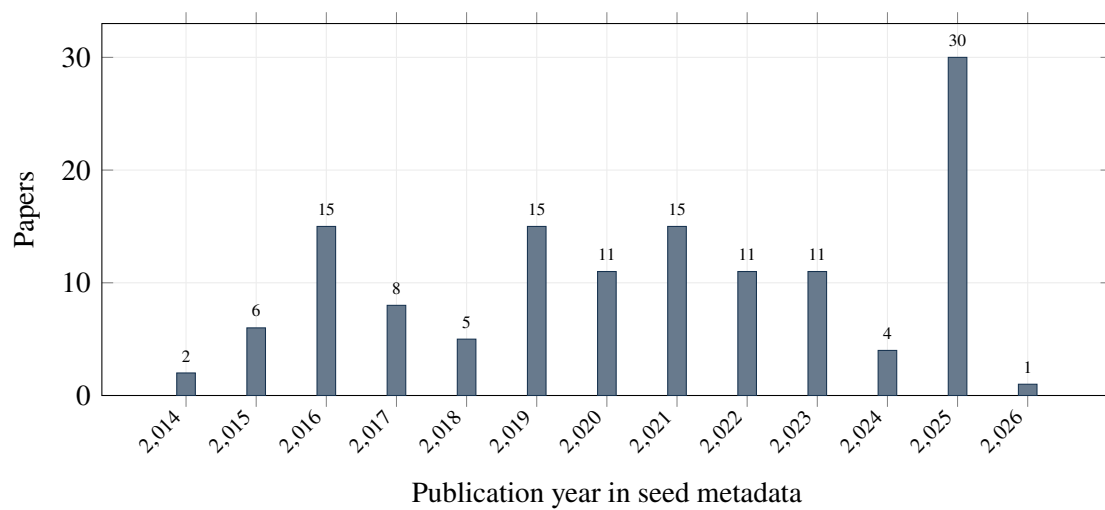


Figure 7.3: Publication-year distribution in the catalog metadata. Missing or uncertain years remain represented in the generated source data rather than being imputed.

Implementation

8.1 Backend and persistence

FastAPI registers public and protected routers; SQLModel maps entities to SQLite. Startup rejects authoritative indexes whose active generation, corpus, configuration, file hash, role, or coverage is invalid. /health reports liveness; /ready separately reports database, retrieval, public-projection, and deployment controls without exposing local paths.

RAG answers retain question, answer, retrieved evidence/citations, provider/model/configuration, corpus/retrieval identity, grounding/support states, warnings, timestamps, and review metadata. Recommendations retain the profile, ranked evidence, gap/suggestion boundary, scope, skills/data/risk/evaluation fields, provenance, and review state. Public serializers omit technical hashes, paths, private notes, and restricted evidence.

8.2 Retrieval modes

Keyword retrieval uses SQLite FTS5 and its `bm25()` rank, with a deterministic token-frequency fallback. Feature hashing maps lower-cased alphanumeric tokens through signed SHA-256 buckets into 256 L2-normalised dimensions. Learned dense retrieval uses the pinned `all-MiniLM-L6-v2` encoder and 384-dimensional L2-normalised vectors. Long chunks are encoded as overlapping 254-token windows, mean pooled, and renormalised. Query/chunk similarity is cosine similarity.

The heuristic hybrid combines normalised keyword/vector scores with metadata, section, evidence, topic, and diversity terms; historical tuning selected keyword 0.10 and dense 0.80 on development MRR. APIs expose four accurately named modes and source/freshness diagnostics. Public retrieval applies the independent publication projection before fusion and rejects stale chunk bytes, generations, or publication state rather than substitute technical rows.

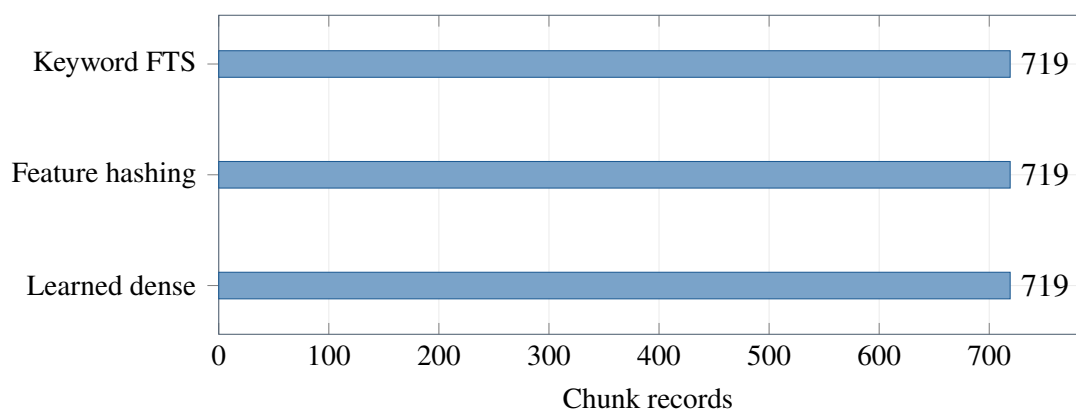


Figure 8.1: Authoritative index coverage. Keyword FTS, feature hashing, and learned dense retrieval each cover all 719 eligible chunks.

8.3 Question answering and citation checks

Figure 8.2 traces Ask. A fixed evidence-coverage rule abstains before composition when a question is not answerable; explicit paper scope handles referential wording. The reproducible provider is offline extraction. Claim decomposition prunes unused citations and records `support_unverified` until atomic review, so locator validity cannot silently become entailment.

The optional Ollama adapter requires an allowlisted tag/digest, rejects pre/post digest drift, and claims generation-time digest verification only when the response reports it; otherwise it records a mutable-tag limitation. Provider failure yields a visible extractive fallback. The historical service was unavailable, so no model-quality or latency comparison is claimed.

The verifier checks identifiers, generation membership, exact current chunk bytes, page locators, fields, and lexical overlap. Its status is structural grounding; AI-silver review assesses support, citation choice, coverage, and abstention separately.

8.4 Thesis Extension Finder

The finder accepts interests, skills, time, project/data constraints, difficulty/topics, retrieval mode, and result count. It groups evidence by paper and scores retrieval 0.30, interest/topic 0.20, skills 0.15, data/time/difficulty 0.10 each, evidence 0.05, and avoided topics -0.20 . These are heuristics, not learned student-success estimates.

Each output separates paper-supported work, explicit/inferred/missing gap, and suggestion; it includes scope, skills, risk, data assumptions, evaluation, citations, and review. Time, access, and feasibility remain candidate-specific unknowns unless sourced. Evidence-only mode stops before generation and supplies the baseline in Fig. 8.3.

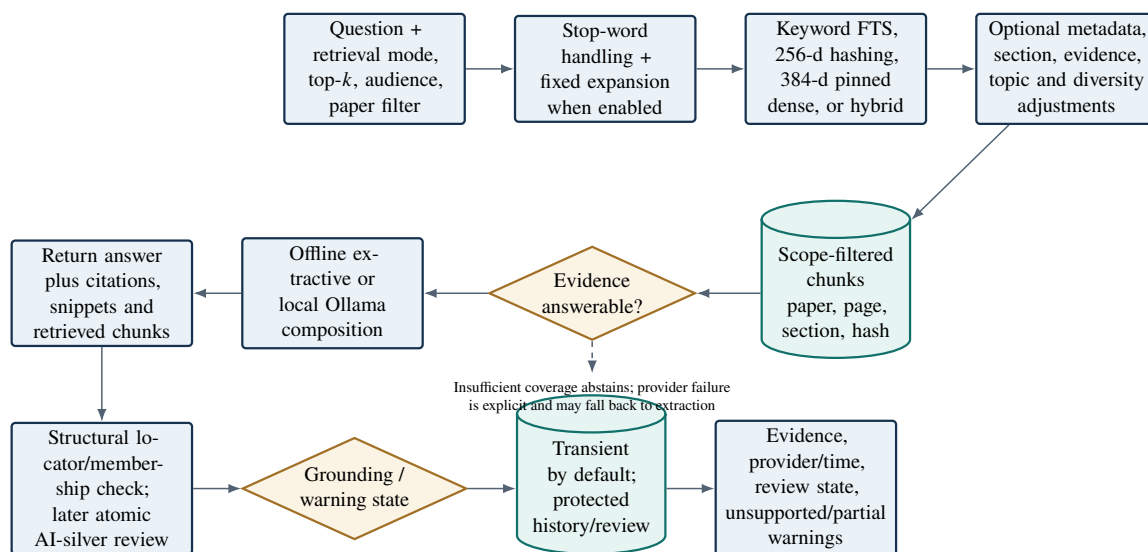


Figure 8.2: Search and RAG workflow. All retrieval modes preserve source locators; structural verification is distinct from claim-level support review.

8.4.1 Finder requirement and failure contract

The Finder’s input contract rejects an empty interest set, bounds list/text lengths, validates enumerated difficulty/project/data values, normalizes weights, and constrains result count. The public POST/api/recommendations/extensions call always uses `persist=False`; protected history/detail calls require a reviewer actor. The browser echoes the submitted profile and mode so a user can detect stale or misapplied constraints. It retains the last successful result while a refresh is in progress and exposes a retry action after failure.

Table 8.1 distinguishes a ranked evidence result from the additional claims in a generated recommendation.

Failure states distinguish over-constraint, unavailable index, no approved public evidence, and no match. Missing gap evidence is not fabricated; partial support retains warnings/citations. Public requests remain transient, expose no technical-scope override, and legacy records without generation identity require reprocessing.

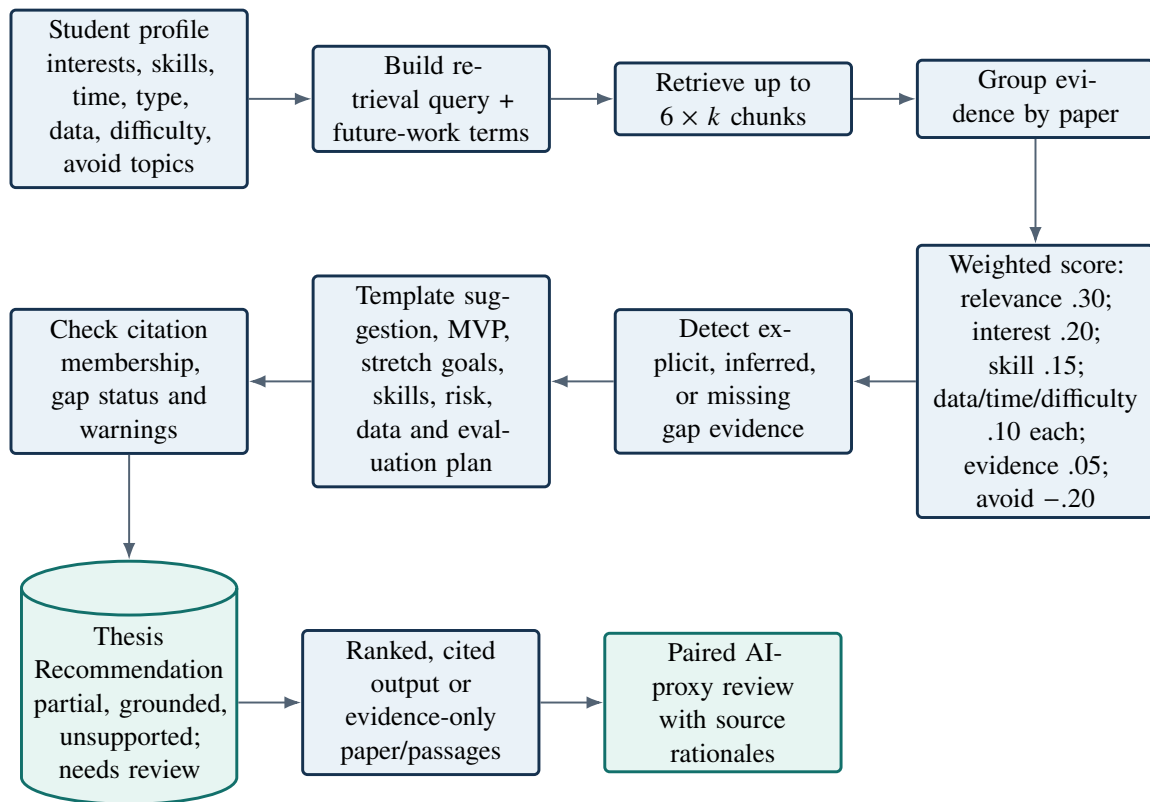
8.5 Topics, authors, artifacts, and review

The explorer uses a controlled 40-label taxonomy with source evidence; a dense prototype provides comparison. Author-topic scores use eligible technical publications, while anonymous serialization applies the public projection and never falls back to hidden relations. Related-paper ranking combines shared topics/authors, metadata, lexical overlap, and optional vectors.

The deterministic artifact generator produces public and technical summaries, contributions,

Table 8.1: Finder output fields, source status, and failure interpretation.

Output	Evidence status	Required user interpretation
ranked paper/passages	retrieved paper/chunk/page evidence	relevant candidate evidence, not project approval
baseline work	source-supported summary of what the paper did	paper fact bounded by cited passages
future work/gap	explicit, inferred, or missing state retained	only “explicit” may be attributed to the paper
suggested extension	system-generated proposal	brainstorming output requiring novelty and feasibility checks
MVP/stretch/skills/data	structured planning fields with assumptions	scope aid; time, access, and mastery remain unverified
risk/difficulty	heuristic categorization plus rationale	warning, not calibrated probability
evaluation plan	proposed metric/baseline/data procedure	incomplete unless dataset, split, baseline, and threshold are defensible
researcher fit	publication-derived topic/author evidence	no availability, endorsement, or supervision claim
grounding/review state	structural verifier plus persisted review metadata	neither state alone establishes human approval

**Figure 8.3:** Recommendation workflow and its 28-profile AI-proxy evaluation. The final suggestion remains advice requiring external feasibility and novelty confirmation.

methods, limitations, future work, suggested extensions, skills, evaluation plans, and text podcast scripts. Fact, inference, and suggestion types are explicit. The Phase 4 review processed every 27 historical RAG answers, 7 recommendation runs, and 14 artifacts. It regenerated evidence-safe artifacts/recommendations where configuration was sufficient; historical RAG configuration was insufficient, so those 27 records were preserved and marked `needs_reprocess`. 48 hash-chained events record the corrections. A second pass changed zero rows, demonstrating idempotence.

8.6 Frontend and evidence communication

The route-based frontend provides all principal public and protected routes. Screens expose source locators, provider/time, grounding/review, freshness, generated notices, and empty/error/retry/offline states. Public emptiness is labelled as an approval/rights boundary; Evaluation separates historical from identity-bound current evidence. Profiles are transient, and automated tests cover semantics, focus, reduced motion, alternatives, and responsive layouts.

Public and protected routes deliberately have different data lifecycles. Ask and Finder send one transient request and do not expose history endpoints. Admin Review is rendered only after a reviewer token is entered; the token is held in module/page memory, attached only to protected requests, and is never written to local or session storage. Reloading clears it. This browser behavior supplements—but does not replace—the backend’s bearer validation and role checks.

8.7 Frontend, API, failure, and test traceability

Table 8.2 connects user-facing requirements to the API and automated evidence. It is intentionally path-specific so that a passing generic page-load test cannot stand in for a citation or authorization contract.

Table 8.2: Requirement–API–view–test traceability for critical public and protected flows.

ID	Requirement/failure state	API and view	Automated evidence
UI-01	Search exposes active mode, snippets, locators, scores, and index warnings; empty/error/retry remain distinct	GET/api/search; Search route	search/index tests; frontend responsible-AI and routing suites
UI-02	Ask identifies generated/transient content, citations, pages, provider, structural grounding, and no-citation state	POST/api/ask; Ask route	backend/tests/test_ask.py; citation and fallback assertions in frontend tests
UI-03	Finder echoes constraints, offers evidence-only/full modes, separates facts/suggestions, and retries failed requests	POST/api/recommendations/extensions; Finder route	test_extension_recommender.py; frontend responsible-AI/E2E scenarios
UI-04	Topic/author relations expose source-derived qualification, aliases, evidence, related papers, and empty states	explorer overview/-topic/author/related endpoints	test_topic_explorer.py; explorer frontend tests
UI-05	Evaluation reports dataset/reviewer boundary and negative findings; missing artefact is “not run” rather than zero	GET/api/evaluation/dashboard; Evaluation route	evaluation API tests; dashboard component assertions

ID	Requirement/failure state	API and view	Automated evidence
UI-06	Anonymous users cannot access review queues/history/mutations; invalid/expired credentials fail visibly	protected recommendation/history and /api/admin/**; protected route shell	test_security_privacy.py; API-client error/auth tests
UI-07	Review supports queue filters, evidence preview, correction, actor-typed transitions, and event history	admin overview/queue/-patch/events endpoints; Admin route	test_admin_evaluation.py; admin component tests
UI-08	AI actors cannot issue human states; event rows cannot be updated/deleted; correction reopens review	admin review/correction endpoints and persistence layer	role, immutability, correction tests in test_security_privacy.py
UI-09	Public question/profile values are request-scoped and not written to history/log output	Ask/Finder transient endpoints; browser memory	non-persistence/redacted-log probes in test_security_privacy.py
UI-10	Keyboard/semantic labels, visible focus, reduced motion, table/chart alternatives, and responsive layouts remain testable	shared components and every primary route	component accessibility suite plus browser checks at 360, 768, 1024, and 1440 pixels

This mapping clarifies the residual human boundary. Automated checks can detect a missing label, hidden focus, unauthorized status code, persistence regression, or absent warning.

They cannot establish whether a student understands the distinction between inferred gap and paper-stated future work, whether a screen reader workflow is efficient, or whether a supervisor considers an extension feasible. Those outcomes belong to a later ethics-approved study and independent accessibility assessment.

The requirement–API–test mapping remains the authoritative behavioural evidence. [Figure 8.4](#) adds two bounded visual records of the implemented evidence boundary. In the Finder state, the fixed synthetic form profile was not sent to the API, diagnostics returned zero publicly searchable papers and chunks, and the action remained disabled. The technical evaluation corpus was not substituted for the empty public projection. In the protected Admin state, a service reviewer can see only aggregate workload counts; the interface identifies the actor type and states that human approval is unavailable. The capture used a physical SQLite snapshot, exact copied indexes, a browser firewall that rejected mutation methods, and before/after source hashes. Its manifest is retained at `thesis/figures/screenshots/governance-capture-manifest.json` and binds the images to clean commit `9a274b82d0f7fa68952369d5ec68565b0a93cbb1`.

The screenshots are therefore illustrative implementation evidence, not substitutes for the route, failure-state, and assistive-technology checks. The executable contracts in Appendix C keep those distinctions inspectable.

8.8 Generated-output governance and deployment boundary

Admin mutations require authenticated roles; production fails closed without HTTPS and an admin actor, and AI actors cannot issue human states. A process-local primitive lock safely spans FastAPI worker-thread handoff, while a database-scoped no-follow file lock serializes review-event mutation/hash/commit across local processes. Neither is distributed consensus. Automatic sync remains disabled until whole-snapshot promotion exists. TLS termination, identity, distributed limiting, monitoring, backup, and incident operations remain deployment responsibilities.

Student Profile

Interests

document processing, information retrieval, and evidence dashboards for a public research archive

Describe research areas or problems; at least one interest is required.

Skills

Python, TypeScript, data visualization

Project Constraints

Available time

semester

Project type

dashboard/visualization

Data constraints

public data preferred

Preferred difficulty

medium

Retrieval Focus

Recommendations

3

Preferred topics

document processing, research discovery

Avoid topics

private student data

Retrieval mode

keyword

Finder output

☒ Full advisor suggestions

☐ Evidence-only ranked papers/passages

0 searchable chunks

0 papers with chunks

Stored recommendation history protected

offline deterministic

Find Thesis Extensions

No papers are approved for public Finder recommendations yet

The public Finder remains disabled until editorial publication, document-rights, extraction, and searchable-access review produce an eligible public corpus. Reviewer-only technical prototype evidence is not substituted into this public route.

(a) Finder with a transient synthetic profile, zero public evidence, and fail-closed recommendation action.

AGGREGATE GOVERNANCE WORKLOAD

Review queues at a glance

Protected

Actor ID

capture-service

Reviewer type

service

Role

reviewer

Human approval

not permitted

These aggregate counts expose no record text. Rights decisions and ingestion remain unavailable unless the server explicitly grants those capabilities.

Papers needing review

134

Missing PDFs

35

Extraction failures

2

Artifacts needing review

0

Ask answers needing review

0

Recommendations needing review

0

Review events

48

Chunks

735

(b) Protected aggregate governance view for a service reviewer without human-approval authority.

Figure 8.4: Current interface evidence for public fail-closed behavior and protected review boundaries. The images demonstrate visible implementation state only; they are not usability evidence, an accessibility conformance result, a human review, a semantic-correctness result, an ethics approval, or a document-rights decision.

Experimental Design

9.1 Overview

Figure 9.1 separates evaluation evidence into offline quality, engineering verification, and external-only validation. It also distinguishes immutable historical v1 diagnosis from the prospective remediation protocol. All datasets store schemas, source/rationale fields, reviewer type, timestamps, review passes, and hashes. Raw full-text prompts and outputs remain local when redistribution would expose third-party passages; sanitized structural records, labels, aggregates, and hashes remain in the release.

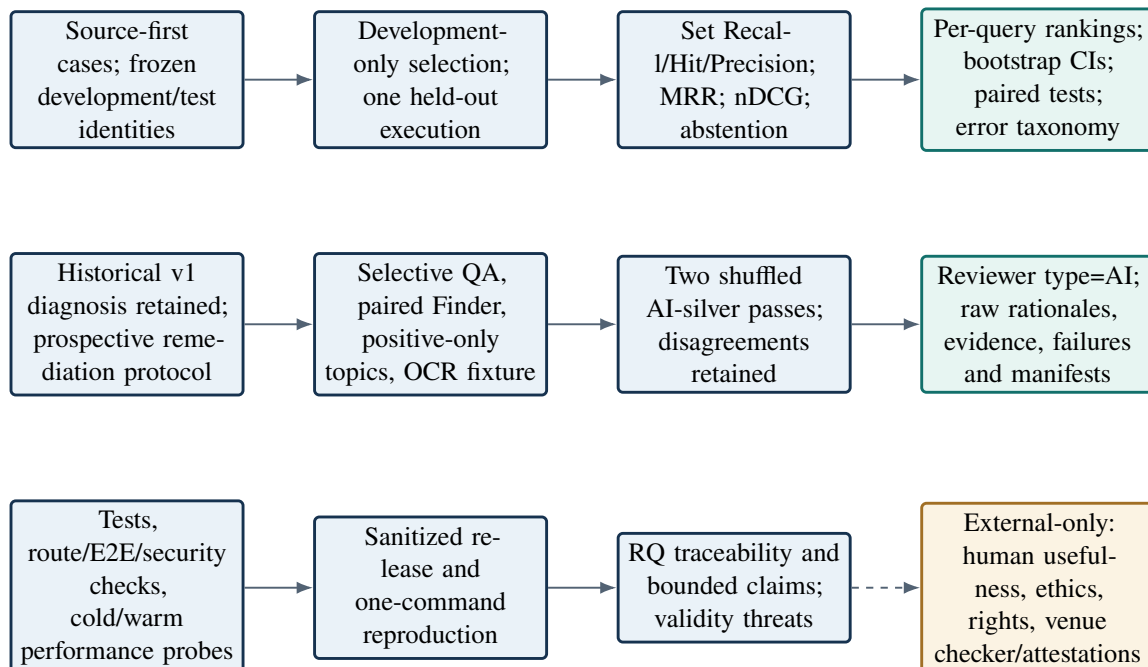


Figure 9.1: Evaluation workflow. Historical diagnosis and prospective AI-reviewed silver evidence are versioned separately from external human or institutional validation.

9.2 Experiment inventory

Table 9.1 lists the historical experiment families. The prospective protocol adds not run selective-QA cases, not run paired Finder profiles, not run known-positive topic cases, and one deterministic OCR fixture. Its declared evidence tier is not available (v2 not run), scope not evaluated, and public-projection execution no; thus it does not establish public behavior, human validation, or entailment.

Table 9.1: Historical v1 experiments, units, and primary outcomes.

Experiment	Cases/unit	Comparison	Primary outcomes
Section quality	40 chunks	predicted versus AI-silver section	accuracy, coverage, macro precision/recall
Retrieval	50 questions (30 dev/20 test)	keyword, hashing, dense, heuristic/tuned hybrid	set Recall, Hit, Precision, MRR, nDCG, abstention, paired tests
QA	50 answers; 400 claims; 81 points	offline extractive outputs against source evidence	claim support, citation correctness/completeness, point coverage, abstention
Recommendation	28 profiles; 84 items/arm	evidence-only versus full finder	proxy relevance, source/claim boundaries, feasibility, scope, risk, sensitivity
Topics	60 papers (36 dev/24 test)	controlled lexical versus dense prototype	multi-label precision, recall, F1, coverage
Author identity	136 rows	schema/source invariants	leakage, collisions, mismatches, overclaims, unresolved candidates
Generated outputs	48 stored records	review pass one versus idempotence pass	status/correction/event integrity
Performance	full-corpus workloads	process-cold versus warm	median, p95, RSS, failures, units
External sanity	3 CC BY papers	fixed lexical discriminators	acquisition/license validation, top-1 format smoke result

9.3 Retrieval configuration

The retrieval dataset contains all eleven specified strata, including four out-of-corpus cases overall. The deterministic split assigns 30 cases to development and 20 to test; the held-out set contains 19 answerable and 1 unanswerable case. All methods use the corpus-04a010207327069a corpus, top 150 chunks per component for frozen candidate pools, a top-50 chunk depth, and top ten unique papers. Unjudged papers count as non-relevant.

Development tuning searched nine keyword/dense weight pairs from 0.10/0.80 to 0.80/0.10, with a 0.42/0.48 implementation reference. The objective was development MRR with fixed secondary rules. The selected tuned configuration used keyword 0.10, dense 0.80, and all heuristic adjustments enabled. Sensitivity tested nearby 0.05/0.85, 0.15/0.75, 0.20/0.70, and 0.30/0.60 weights. Individual ablations removed one component from the selected hybrid; cumulative ablations added components in the documented order.

9.4 Historical v1 QA review procedure

The offline extractive answerer generated all 50 historical reviewed answers. Each atomic corpus claim was linked to inline source IDs and returned citation records. The AI reviewer performed a support-first creation pass and a shuffled contradiction-focused verification pass; the verification label was final, with disagreements retained. The metrics use strict treatment of partial labels. Confidence intervals resample whole QA cases 10,000 times, preserving within-case claim correlation. They do not include AI-label uncertainty.

An Ollama comparison was attempted at the configured local endpoint on port 11434. CLI version 0.21.2 was present, but version, tags, and process endpoints refused connections. No installed model, digest, quantisation, response, or latency could be verified, so the comparison is reported as unavailable rather than approximated.

9.5 Historical v1 recommendation, topic, and review protocols

The 28 synthetic profiles span one-month, two-week, and semester timelines; software, ML, analysis, dashboard, literature-support, and other project types; public/synthetic/no-external/supervisor data constraints; easy/medium/hard preferences; and avoid topics. Both arms return three items. Review criteria are applied to every item in two shuffled AI passes. The paired relevance contrast and sensitivity results resample profiles, not individual recommendations.

The topic dataset contains 36 development and 24 test papers, covers all controlled labels across the full dataset, includes 51 multi-label cases and four other/unknown cases. The dense threshold (0.20) was selected by development micro-F1, with macro-F1 and higher threshold as tie-breakers. Test data were not used for selection. The author audit checks only source-backed invariants; candidate merges are preserved for external identity confirmation.

Generated-output review covers every stored RAG answer, recommendation run, and paper artifact. The procedure records rationale, citation locators, corrections, actor

type, and chained event hashes. Configuration-incomplete historical outputs remain `needs_reprocess`; they are not deleted or silently regenerated.

9.6 Performance and reproducibility protocol

The full benchmark uses three process-cold and three warm repetitions at concurrency one. Each workload runs in an isolated database/index/output context and retains raw timing and bounded errors. The operating-system cache is not flushed. A linearly interpolated p95 over three observations is coarse and is reported only as a local baseline. The one-command reproduction path verifies dependencies, rebuilds allowed derived state, runs evaluations and tests, compiles both manuscripts, performs PDF preflight, builds the sanitized release, and emits hashes. Exact full-corpus reruns require lawful access to local PDFs and the pinned dense model.

Results

10.1 RQ1: corpus, extraction, and index integrity

The catalog contains 134 records. 98 PDFs were available; 96 papers passed the frozen inclusion rules, 36 had no eligible full text, and 2 were excluded for suspected PDF/title mismatch. The raw database contains 735 chunks; 719 are eligible, and 199 of those retain an unknown section. SQLite quick check returned ok and foreign-key checking found no violations.

The 40-case AI-silver section review produced exact accuracy 0.900, labeled coverage 0.900, macro precision 0.892, and macro recall 0.904. The unknown-section share remains material, so section-aware ranking is a fallible signal.

Keyword, feature-hashing, and learned-dense indexes each cover 719/719 eligible chunks (Fig. 8.1). Their validated manifests share corpus snapshot corpus-04a010207327069a, ordered chunk/source identities, and the authoritative role. The prior partial-index/stale-status defect is therefore closed for the frozen technical corpus. The bounded conclusion for RQ1 is that complete derived-state coverage is achieved for the eligible subset, while 36 no-text and 2 mismatch records remain outside it and 199 eligible chunks lack a confident section label. This does not make those technical records public: the audited public searchable projection is empty pending human approval, rights clearance, extraction approval, and current public-generation promotion.

10.2 RQ2: historical v1 held-out retrieval

Table 10.1 reports held-out estimates. Keyword retrieval had the highest point estimate on all four headline answerable-query measures. Learned dense achieved MRR 0.939, set Recall@3 0.916, and set Recall@10 0.989, compared with keyword values 0.947, 0.939, and 1.000. Feature hashing had lower point estimates. The tuned hybrid improved over feature

hashing but did not exceed keyword or learned dense on these point estimates.

Table 10.1: Held-out retrieval results on 19 answerable questions. Recall@3 intervals are 10,000-replicate query bootstrap 95% CIs.

Mode	Set R@3 (95% CI)	Set R@10	MRR	nDCG@10	Unans. FP
Keyword	0.939 [0.866, 1.000]	1.000	0.947	0.958	1/1
Feature hashing	0.654 [0.443, 0.851]	0.704	0.588	0.582	1/1
Learned dense	0.916 [0.821, 1.000]	0.989	0.939	0.939	1/1
Heuristic hybrid	0.795 [0.611, 0.947]	0.917	0.813	0.829	1/1
Tuned hybrid	0.856 [0.707, 0.974]	0.925	0.860	0.871	1/1

The tuned-minus-keyword MRR contrast was -0.088 (95% CI -0.237 to 0.044 ; exact paired randomisation $p = 0.375$, experiment-family Holm-adjusted $p = 1.000$). Tuned-minus-dense was -0.079 (95% CI -0.175 to 0.000 ; adjusted $p = 1.000$). Tuned-minus-feature-hashing was $+0.272$ (95% CI 0.026 to 0.518), but the paired $p = 0.061$ and family-adjusted $p = 1.000$ did not reject the null. No corrected comparison demonstrated superiority.

Cumulative ablations were descriptive because they were examined after configuration selection. Adding topic adjustment produced set Recall@3 0.916, Recall@10 1.000, MRR 0.947, and nDCG@10 0.952; adding diversity did not change those aggregates. The error taxonomy flagged 1 extraction/unknown-section case, 2 intent/ranking cases in which relevant papers fell below rank ten, and 1 unanswerable corpus-absence case. All five modes returned a positive result for the one held-out unanswerable query. This single case is too small for a stable rate; it shows only that none of the evaluated configurations abstained on that case. The RQ2 conclusion is that keyword had the highest held-out point estimates, learned dense was lower on every headline point estimate, and no corrected comparison established tuned-hybrid superiority.

10.3 RQ3: historical v1 claim support, citations, completeness, and abstention

Table 10.2 separates support from completeness. Of 400 checkable claims, 398 were supported, 2 partially supported, and 0 unsupported. Strict supported-claim rate was 0.995 (95% CI 0.987–1.000). Every checkable claim carried a citation, but only 250/400 claim–citation links were fully correct: 0.625 (95% CI 0.564–0.691). Strict answer-point coverage was 11/81, or 0.136 (95% CI 0.067–0.211). Returned citation utilisation was 0.592.

The error analysis found incomplete answers in 44/50 cases, off-topic retrieval in 42, within-paper evidence misses in 14, full answer-point omissions in 13, answer sentence-selection misses in 12, incorrect citations in 4, and false-positive answers for all 4 unanswerable cases.

Table 10.2: AI-assisted QA review over 50 offline extractive answers. Partial labels do not count in strict estimates.

Metric	Estimate	95% CI	Interpretation
Supported-claim rate	0.995	[0.987, 1.000]	local claim/source support
Citation correctness	0.625	[0.564, 0.691]	correct claim–citation links
Citation completeness	1.000	[1.000, 1.000]	citation presence only
Answer-point coverage	0.136	[0.067, 0.211]	strict expected-content coverage
Answerability accuracy	0.920	[0.840, 0.980]	answer/abstain decision
Unanswerable abstention	0.000	[0.000, 0.000]	0/4 cases abstained

Thus, source-traceable extraction produced few locally unsupported claims in this sample, but it did not deliver complete answers or correct abstention. The RQ3 conclusion is a negative/mixed result: high local claim support coexists with weak citation choice, low answer-point coverage, and failed abstention, so structural citation presence overstates end-to-end answer quality.

10.4 RQ4: historical v1 recommendation proxy evaluation

Across 28 profiles, each arm produced 84 reviewed items. Evidence-only AI-proxy paper relevance was 0.631 (95% CI 0.524–0.726); full-finder relevance was 0.655 (95% CI 0.560–0.750). The paired full-minus-baseline estimate was 0.024 (95% CI -0.024–0.071), so improvement was not demonstrated.

All 84 full-finder outputs passed source-fidelity, fact/future-work/gap/suggestion separation, novelty-caution, MVP, stretch-goal, risk, and required-skill checks. All 84 feasibility judgments were partial because the source evidence could not validate time, skill mastery, or data access. Evaluation-plan quality passed in 72 cases and was partial in 12. The two AI passes agreed exactly at 0.994 over 1092 criterion judgments; this is same-reviewer consistency, not human reliability. Weight sensitivity showed the interest term was the most disruptive removal, while most $\pm 25\%$ changes retained high top-three overlap. This component of RQ4 finds that claim separation is structurally strong, but relevance improvement is small and inconclusive and feasibility remains external.

10.5 RQ5: historical v1 topics, authors, and generated review

The held-out topic results in Table 10.3 show a precision–recall trade-off. The controlled lexical method achieved higher precision and F1; the dense prototype achieved higher recall

and coverage but produced more semantic overreach. The controlled vocabulary was retained for its precision, inspectability, and editability.

Table 10.3: Held-out multi-label topic results on 24 AI-silver paper cases.

Method	Micro precision	Micro recall	Micro F1	Coverage
Controlled lexical	0.652	0.484	0.556	0.875
Dense prototype	0.417	0.694	0.521	0.958

Aggregate topic metrics are supported unevenly across labels. Only 26 of the 40 vocabulary labels had a positive held-out example; 14 labels had no gold support and cannot contribute an observed positive-label recall. Table 10.4 reports representative confusion counts selected before manuscript drafting to include strong, weak, broad, and domain-specific behavior.

Table 10.4: Representative held-out controlled-lexical topic errors. Precision/recall are undefined in principle when a denominator is zero; generated zero values are retained only for table consistency.

Label	TP	FP	FN	Precision	Recall
Agriculture	2	1	0	0.667	1.000
AI	0	0	4	0.000	0.000
Clustering	3	1	0	0.750	1.000
Energy	1	0	2	1.000	0.333
Networks	0	2	2	0.000	0.000
RAG	2	0	0	1.000	1.000

The author audit found zero excluded-evidence leakage, alias collision, eligible authorship mismatch, excluded-paper topic link, or overclaim-output violation. It retained 13 possible same-person pairs for external confirmation and derived 313 author-topic links from eligible publications. Consequently, topic/author exploration is inspectable but not author-validated.

Generated-output review processed all 48 stored records. 21 artifacts/recommendation records were regenerated or verified and marked `ai_reviewed`. 27 historical RAG answers lacked sufficient generation configuration for safe reproduction; they remain preserved as `needs_reprocess`. The first pass created 48 hash-verified audit events and changed 48 records; the second pass created zero events and changed zero records. The RQ5 conclusion is that discovery outputs and corrections are inspectable and AI review is attributable/idempotent, while human approval, complete expertise, endorsement, and unresolved identity merges remain external.

10.6 Prospective remediation results for RQ3–RQ5 and OCR

Package not completed or validated (not available (v2 not run)) binds protocol not run, datasets not run, environment not run, source not run, snapshot not run, and not run keyword state. These identify an execution; they do not prove correctness.

Selective QA over not run cases produced answerability precision/recall not run (not run) / not run (not run), false-positive rate not run (not run), and abstention not run (not run). Locator precision was not run (not run), citation completeness not run, and point coverage not run. These test repaired contracts, not entailment.

Across not run profiles, evidence/full Hit@3 was not run (not run) / not run (not run), with change not run (not run). Corresponding MRRs were not run and not run (change not run). Template, fact/suggestion, unknown-time, and constraint contracts were not run, not run, not run, and not run. These are contract/proxy evidence, not usefulness, novelty, feasibility, or supervisor fit.

Across not run positive-only cases, recall was not run (not run) and case coverage not run; not run predictions remained unadjudicated, so precision/F1 and public behavior are not established. OCR fixture status not run yielded character/word error not run/not run, exact match not run, and repeatability not run; it is not scanned-corpus accuracy.

10.7 RQ6: reproduction, performance, and external sanity

The final benchmark results are generated from the strictly validated artifacts/phase6/performance/performance_full_results.json and summarised in Table 10.5. The benchmark uses 3 cold and 3 warm repetitions at concurrency one on AMD Ryzen 7 5800X 8-Core Processor with 3.8 GiB available memory under WSL2. Across all 17 workloads, 102 cold/warm samples and all 102 RSS observations passed validation with 0 failures; maximum observed process RSS was 751.5 MiB. The clean benchmark source commit is 5ccf22ec and the provenance-hash prefix is 61378b9aa78a.

The external sanity harness mapped three CC BY JATS/XML articles into the production chunker contract and achieved 3/3 fixed lexical top-one matches; it did not validate the main PDF-ingestion pipeline or cross-domain retrieval quality.

The sanitized release scanner excludes PDFs, SQLite, indexes, local paths, secrets, raw/full extracted text, interface screenshots, private prompts, and raw Europe PMC XML. It verifies safe archive members and per-file/archive SHA-256 checksums. The final engineering,

Table 10.5: Selected full-corpus Phase 6 performance results. Times are seconds; p95 is a coarse interpolation over three repetitions.

Workload	Cold median	Cold p95	Warm median	Warm p95
Dense indexing	145.005	145.020	66.169	134.066
Keyword retrieval	9.809	10.126	8.091	9.526
Dense retrieval	84.487	86.265	12.215	78.342
Hybrid retrieval	93.355	93.616	20.326	86.385
Offline answer	83.355	83.736	9.842	75.800
Recommendation	83.604	85.675	10.482	76.668
ASGI endpoint set	79.606	81.178	3.948	72.397
Frontend route set	12.496	12.807	12.463	12.631

document, preflight, release, and reproduction outcomes are recorded in thesis/PROJECT_EVIDENCE.md. The bounded RQ6 conclusion is local reproducibility and operational characterization under the documented configuration, privacy-minimizing release, and limited JATS/XML format compatibility. Production scale, main-pipeline external PDF compatibility, and unrestricted corpus reproduction are not established.

10.8 RQ7: evidence-interface and governance contracts

The public runtime probe exercised statistics, browse/detail, search diagnostics, transient Ask, transient Finder, explorer, evaluation, and readiness paths against a copied database or read-only source state. Because no paper met the independent publication-and-rights predicate, evidence-bearing public routes returned empty/degraded states and never substituted the technical evaluation corpus. The probe also confirmed that public Ask/Finder calls do not add history records and that anonymous Admin access is denied. Protected/local evaluation separately exercised the generation-linked technical corpus. Backend security tests covered bearer/role enforcement, production fail-closed configuration, correction-driven state reopening, AI-versus-human transition restrictions, public serialization, path/URL limits, redacted logging, and review-event immutability.

The frontend production build and route/component/browser suites exercised the Dashboard, Paper Browser/Detail, Search, Ask, Finder, Topic/Author Explorer, Evaluation, and protected Admin shell. The critical assertions were not only route visibility. Ask exposes its generated/transient notice, provider, structural-grounding boundary, citations, pages, and no-citation/error states. Finder echoes the submitted profile, offers evidence-only/full modes, separates paper facts from suggestions, and preserves a retryable failure. Evaluation displays reviewer/dataset boundaries and missing-result states. Admin requires a page-memory bearer token before protected data loads and renders queues, citation previews, typed status controls, corrections, and event history.

Automated accessibility regressions cover semantic headings/landmarks, programmatic labels, keyboard focus, reduced motion, table/chart alternatives, and responsive routes at the documented viewport set. These checks passed in the recorded candidate, but they are conformance aids rather than a WCAG audit. No participant or assistive-technology study was conducted. The RQ7 conclusion is therefore that evidence, uncertainty, transient-input, authorization, failure-recovery, and correction contracts are implemented and testable. Comprehension, trust, task success, accessibility conformance, and production operational governance remain unestablished.

Discussion

11.1 Traceability is an engineering property, not an accuracy score

The platform now preserves paper, chunk, page/section, extraction/chunk/index generation, provider/configuration, warning, timestamp, and review state through its principal workflows. Validated active-generation pointers eliminate the earlier ambiguity between index files and stale database flags. These are meaningful engineering results because they make an output auditable and stale derived state detectable. They do not make the output correct.

The historical QA evaluation illustrates the distinction. Extractive composition largely avoided unsupported statements by copying supported passages, yet claim–citation choice and answer-point coverage were much weaker. It selected off-topic evidence and omitted requested information even when every sentence carried a locator. The prospective run separately tests the repaired selective-response and exact-locator contracts; its citation precision is not run and answer-point coverage not run. Neither measure turns a matching locator into entailment. Traceability reduces the cost of diagnosis; it does not substitute for diagnosis.

11.2 Simple lexical retrieval remained competitive

Keyword retrieval produced the highest held-out MRR and nDCG@10 point estimates. Learned dense was close, but the study was not designed as an equivalence or non-inferiority test. The tuned hybrid ranked below both and did not yield a corrected significant gain even over feature hashing. This agrees with the broader lesson of heterogeneous retrieval benchmarks: learned representations are not automatically superior for every corpus [15].

Several features may explain the keyword result without establishing causality. The labora-

tory questions and paper titles share specialised vocabulary; relevant-paper labels may favour explicit lexical matches; the corpus is small; and the candidate pool is already rich. The post-selection ablation showing a stronger cumulative configuration is hypothesis-generating, not an invitation to retune on the held-out set. A future release needs a larger preregistered test partition and hard paraphrase cases.

11.3 Selective response is improved but narrowly evaluated

Historical v1 retrieval returned a positive result on its lone held-out unanswerable case, and the historical answerer failed to abstain on every QA unanswerable case. The repaired fixed coverage gate changes that behavior prospectively: answerability precision is not run with false-positive rate not run and abstention rate not run. This is stronger evidence of the implemented selective-response contract, but the held-out set remains small and source-derived. A later protocol should expand adversarial and out-of-corpus cases and report risk jointly with retained answer coverage.

11.4 Recommendation structure is stronger than recommendation validation

The extension schema separates baseline evidence, explicit versus inferred gaps, system suggestions, MVP/stretch scope, risk, skills, data assumptions, and evaluation plans. Historical review found every feasibility judgment partial and an inconclusive relevance difference. The prospective paired test measures ranking and contract behavior: its Hit@3 change is not run (not run), while fact/suggestion separation is not run and unknown-time retention not run. These values do not establish student usefulness or real feasibility. The full Finder remains a structured brainstorming aid, and evidence-only mode lets a cautious user stop before generated advice.

11.5 Controlled discovery favors inspectability

Historical topic comparison exposed a practical trade-off: the dense prototype increased coverage/recall but reduced precision, while controlled rules were deterministic and editable. The prospective positive-only set instead asks whether known positives are recovered, yielding recall not run while leaving not run additional predictions unadjudicated. It cannot confirm precision or expertise. The remaining 13 identity candidates show why automatic name merging is intentionally conservative. Publication-derived relations should remain qualified even after identity resolution.

11.6 Review and governance as first-class data

Append-only AI review events close the earlier empty-governance state and demonstrate deterministic idempotence. The design does not conflate that activity with human approval: reproducible outputs are AI-reviewed, while configuration-incomplete historical answers remain needs-reprocess. Authentication, actor roles, chain hashes, and protected status transitions make future human review attributable. A deployed institution still needs operational ownership, identity lifecycle, correction response times, and legal/privacy decisions.

11.7 Interface conformance is not human validation

The requirement-API-view-test trace shows that implemented public and protected flows preserve warnings, request-lifecycle boundaries, authorization, and correction state under tested conditions. At this snapshot the strict public projection is empty, which makes fail-closed behavior testable but provides no evidence about a populated public experience. A semantic label can exist while remaining confusing; keyboard focus can be visible while a screen-reader workflow remains inefficient; and a Finder warning can render while a student still over-trusts the suggestion. RQ7 therefore closes an engineering traceability gap without converting automated conformance into usability, accessibility, trust, or educational evidence.

11.8 Reproducibility with rights constraints

An exact full-corpus replication requires lawful access to the third-party papers; publishing their text would improve convenience by violating the evidential and rights boundary. The sanitized release therefore favours computational transparency over unrestricted data redistribution. It includes code, schemas, labels, sanitized results, prompts/configurations, hashes, code-native figure sources, and instructions, while requiring recipients to reacquire permitted documents. Interface screenshots are excluded because rendered source/answer text cannot be field-redacted reliably. The external sanity harness mapped open JATS/XML documents into the production chunker's extraction contract. It did not exercise the primary PDF-ingestion path, and a few life-science articles do not establish domain transfer.

Limitations, Ethics and Threats to Validity

12.1 Technical and empirical limitations

Only 96 of 134 records enter full-text experiments; 36 lack text and 2 mismatches require review. Extraction is heuristic, corpus OCR was not run, and the fixture is not scanned-corpus accuracy. 199 chunks lack sections, metadata are incomplete, 13 identities remain unresolved, and the empty public projection supplies no populated-public evidence.

Historical v1 retrieval has 50 AI-silver cases and 20 held out, including one unanswerable; pools may omit relevance, unjudged documents count negative, and ablation was post-selection. Prospective sets are also small, source-derived and technical-only. All intervals omit label and corpus-selection uncertainty.

QA uses one extractive answerer and one AI-review process; high historical support partly reflects copying, and prospective results cover only frozen cases. Ollama was unavailable, so no generative comparison exists. Recommendation/topics remain AI-proxy/silver, not user judgments. Single-machine concurrency-one performance is not capacity testing.

12.2 Ethics, permissions, and responsible AI

TTLAB explicitly authorized the project and use of the laboratory corpus. This fact does not establish research-ethics-board approval or exemption, ownership of third-party publications, or blanket redistribution permission. No human participants were recruited; no satisfaction, usability, educational benefit, or supervisor-approval claim is made. If a later human study is undertaken, institutional ethics determination, informed consent, recruitment, retention, compensation, and withdrawal procedures must precede data collection.

Potential harms include source misrepresentation, inappropriate thesis advice, expertise misattribution, copyright overexposure, privacy leakage from project profiles, malicious documents, and unauthorised review changes. Controls include source locators, fact/inference/suggestion labels, AI/human review separation, warnings, evidence-only mode, transient profiles, authenticated mutations, append-only event hashes, PDF/downloader limits, and a sanitized release. These controls reduce risk; they do not amount to institutional approval or human verification.

Codex/LLM assistance was material in coding, evidence inspection, test and evaluation preparation, AI-silver labeling, analysis, and manuscript editing. Reviewer IDs, prompts or prompt hashes, seeds, and limitations are persisted. The exact underlying Codex model digest was not exposed by the runtime and was not invented. This disclosure should be reconciled with the University's current submission policy by the author/institution before formal submission.

12.3 Construct validity

Set Recall@ k , Hit@ k , MRR, nDCG, citation correctness, citation completeness, supported-claim rate, answer-point coverage, and abstention are defined separately. "Grounded" in operational records denotes structural checks, not entailment. Topic score denotes controlled or prototype label evidence, not topic truth. Recommendation score denotes heuristic fit, not usefulness, novelty, or feasibility. AI-review consistency is not human inter-rater reliability. Index coverage and test counts are engineering properties, not effectiveness.

12.4 Internal validity

Development/test separation and prospective pre-execution case/split freezing reduce leakage. Nevertheless, source-derived labels came from one AI-assisted process and may reflect corpus vocabulary; the same AI reviewer verified them. Error tags are non-causal, and WSL load/caches affect performance.

12.5 External validity

The system and controlled taxonomy are tailored to one laboratory, English-language records, and a small local corpus. Archive markup, disciplinary terminology, PDF quality, and user goals will differ elsewhere. The three-document Europe PMC check covers only CC BY life-science JATS/XML and trivial lexical queries. It cannot support claims about other institutions, languages, domains, or large corpora. No human study supports

generalisation to student decision-making or public understanding.

12.6 Reproducibility validity

Code, structural results, model/environment identities, seeds, and manifests are versioned and bind active index generations. Exact replication still needs lawful sources, model availability and compatible tools. Restricted-raw hashes attest runner observations, not independently inspectable content; open analytical and conditional full-corpus reproduction remain distinct.

12.7 Security and deployment limitations

The repository provides a fail-closed production configuration boundary, but it is not a hosted service. TLS, reverse proxy, identity-provider lifecycle, a distributed rate limiter, central monitoring, backups, incident response staffing, legal notices, accessibility conformance assessment, and operational correction processes remain deployment-specific. Scheduled corpus synchronization is deliberately disabled until whole-snapshot atomic promotion exists.

The current Tectonic thesis PDF is untagged. Semantic source structure, text alternatives in figure captions/descriptions, bookmarks, selectable text, and embedded fonts improve portability, but they do not establish PDF/UA conformance. A tagged-PDF-capable toolchain and independent accessibility inspection are required before making that claim. External IEEE PDF Express/Checker validation and any venue/institutional declarations likewise require credentials and authorized human attestations.

Conclusion and Future Work

This thesis designed and evaluated a source-traceable research-intelligence platform for a laboratory publication corpus. Its contribution is not a single chatbot: it is an evidence-preserving path from corpus eligibility through retrieval, citation-bearing QA, thesis-extension support, controlled discovery, generated artifacts, evaluation, and attributable review. The results answer the seven research questions as follows.

- RQ1** The frozen eligible technical corpus and its keyword, feature-hashing, and learned-dense representations are complete under recorded manifests and generation links. Unavailable records, mismatches, unknown sections, and the empty independently approved public projection remain explicit rather than silently repaired or bypassed.
- RQ2** Historical v1 keyword retrieval has the strongest held-out point estimates; learned dense remains competitive, while the tuned hybrid does not demonstrate corrected superiority. The small AI-silver set supports bounded comparison, not equivalence or universal ranking claims.
- RQ3** Historical atomic review exposed weak citation choice, low answer-point coverage, and failed abstention despite high local support. The prospective selective run reports answerability precision not run and exact-locator citation precision not run; these are technical AI-silver results, not entailment or human usefulness.
- RQ4** The Finder preserves candidate-specific fact/suggestion and unknown-feasibility boundaries. Its prospective paired Hit@3 change is not run with interval not run. It remains a structured brainstorming aid, not validated thesis advice.
- RQ5** Controlled topic labels, conservative author identity, and actor-typed review make discovery outputs inspectable. Prospective known-positive recall is not run, but additional predictions remain unadjudicated; no comprehensive expertise, endorsement, public behavior, or human approval is established.
- RQ6** Versioned manifests, a sanitized rights-aware bundle, repeatable local builds, and

an open-XML sanity check support bounded reproduction. They do not establish primary-PDF transfer, concurrent production capacity, or broad domain generalization.

- RQ7** Public and protected interface tests verify evidence display, transient profiles, authorization, correction history, failure states, responsive layouts, and selected automated accessibility contracts. They do not establish comprehension, satisfaction, WCAG or PDF/UA conformance, or assistive-technology usability.

The principal conclusion is deliberately restrained: the artifact makes research-intelligence outputs easier to inspect, reproduce, and correct, but it is not a human-validated research advisor. The most important negative finding is the gap between traceability and utility: a response can carry current source identifiers while selecting the wrong evidence or omitting requested points. The strict public gate adds a complementary lesson: honest emptiness is preferable to presenting an authorized technical corpus as if every paper had been individually approved for public delivery.

13.1 Prioritised future work

1. **Calibrate selective response.** Expand unanswerable and adversarial cases, select evidence/score thresholds on development data, and report risk jointly with retained coverage.
2. **Improve evidence selection.** Add answer-point-aware reranking and within-paper passage retrieval, then repeat citation-link and completeness review without tuning on the present test set.
3. **Run independent human studies.** After institutional ethics determination, recruit students, supervisors, researchers, and public users to evaluate relevance, comprehension, feasibility, trust, accessibility, and workflow utility.
4. **Resolve data, rights, and identity gaps.** Acquire only permitted missing PDFs, obtain paper-level publication/rights decisions, externally verify mismatch documents and candidate author merges, and evaluate OCR on a real scanned-document subset.
5. **Replicate externally and at scale.** Freeze larger independent test sets, evaluate additional authorized laboratories, disciplines, languages, and PDF formats, and run concurrent load/endurance tests in a production-like deployment.
6. **Strengthen governance.** Integrate institutional identity, retention/deletion policy, correction and appeal processes, monitoring, backups, incident response, and independent security and accessibility assessment.

Research-Question Traceability Matrix

Table A.1: Complete connection from each research question to method, evidence, result, and bounded conclusion.

RQ	Method	Authoritative evidence	Principal result	Bounded conclusion
1	eligibility/identity audit; generation reconciliation; extraction/OCR fixture; manifest and public-projection validation	Phase 1 JSON; database integrity; section cases; active index generations; remediation OCR output	technical representations are complete; mismatch/unknown states persist; public searchable projection is empty; fixture CER is not run	the technical corpus is fail-loud; no public approval, scanned-corpus OCR accuracy, or unavailable-text recovery is implied
2	historical v1 development/test comparison; tuning; ablation; bootstrap and paired tests	immutable per-query rankings; summaries, statistics, errors, tuning and sensitivity outputs	keyword has the highest historical held-out point estimates; corrected comparisons do not show tuned-hybrid superiority	bounded post-selection diagnostic behavior is measured; neither equivalence nor broader-corpus superiority is established
3	historical atomic diagnosis plus prospective source-first selective-QA execution	historical claim/point decisions; remediation raw structural outputs, review passes, disagreements, metrics, and macros	historical citation/coverage/abstention failures persist; prospective answerability precision is not run and locator precision not run	repaired technical contracts are measured; source membership does not establish entailment, complete answers, or human usefulness

RQ	Method	Authoritative evidence	Principal result	Bounded conclusion
4	historical proxy diagnosis plus prospective paired evidence-only/full Finder execution and sensitivity	historical criterion decisions; remediation paired rankings, contract checks, metrics, and macros	historical improvement is inconclusive; prospective Hit@3 change is not run and fact/suggestion separation not run	inspectable ranking/advice contracts are measured, not novelty, feasibility, supervisor fit, or student benefit
5	historical closed-world topic diagnosis; prospective known-positive evaluation; author/output audit	historical topic errors; remediation predictions/metrics; identity audit; review events/idempotence	prospective known-positive recall is not run; extra predictions remain unadjudicated; 13 identities remain unresolved	discovery and review are inspectable, not topic precision, comprehensive expertise, endorsement, identity certainty, public behavior, or human approval
6	repeated benchmark; full/quick reproduction; release scan; external open-corpus check; document preflight	Phase 6 raw results; reproduction/release manifests; 3 acquisition records; PDF checks	concurrency-one operation is characterized; external XML maps into the chunker and gives 3/3 fixed lexical top-one matches; restricted PDFs remain excluded	local reproducibility and limited format compatibility are supported, not production capacity, primary-PDF transfer, or broad external validity
7	requirement-API-view-test trace; public privacy probe; role/authentication, failure, responsive, and accessibility tests	route/API contracts; runtime probe; backend security and frontend component/E2E evidence	transient public requests, protected history/mutations, evidence states, failure recovery, and automated accessibility contracts pass their specified checks	implementation conformance is supported; comprehension, satisfaction, WCAG/PDF-UA conformance, assistive-technology usability, and production readiness remain external

Table A.1 is the authoritative claim map. Inventory evidence supports RQ1, immutable historical v1 and prospective AI-silver experiments support bounded RQ2–RQ5 observations, reproduction evidence supports RQ6, and interface tests support RQ7. None substitutes for human validation, an institutional decision, paper-level rights approval, or production assessment.

Reproducibility Guide

Run commands from the repository root on Linux/WSL. The exact code revision, working-tree state at execution, environment, input hashes, and output hashes are embedded in phase and reproduction manifests. The retained historical-v1 snapshot is corpus-04a010207327069a; the prospective run binds its rebuilt technical snapshot separately as not run.

B.1 One-command paths

```
make reproduce-quick # code/sanitized evidence path
make reproduce # full authorized-corpus path
make release # deterministic sanitized bundle
```

The full path requires a lawful local database/PDF corpus and the pinned dense model. It imports/repairs metadata, reconciles source generations, extracts/chunks, validates or rebuilds immutable index generations, executes evaluations and performance workloads, runs tests/builds, generates manuscript evidence, compiles paper and thesis, performs local PDF pre-flight, builds the release, and emits a final manifest/checksums. The quick path verifies distributable code and sanitized evidence without claiming to recreate restricted inputs.

B.2 Evidence and manuscript generation

```
PYTHONPATH=backend .venv/bin/python thesis/scripts/collect_evidence.py
PYTHONPATH=backend .venv/bin/python thesis/scripts/runtime_probe.py
PYTHONPATH=backend .venv/bin/python thesis/scripts/
    generate_manuscript_assets.py
# With the backend on :8000 and frontend on :4173:
PYTHONPATH=backend .venv/bin/python \
```

```

-m app.reproducibility.database_snapshot \
--source data/papers.db \
--target /tmp/ttlab-capture-XXXX/papers.db \
--evidence /tmp/ttlab-capture-XXXX/database-snapshot.json
TTLAB_CAPTURE_ISOLATED=1 \
TTLAB_SCREENSHOT_CAPTURE_SET=governance \
TTLAB_SCREENSHOT_RUNTIME_ROOT=/tmp/ttlab-capture-XXXX \
TTLAB_SCREENSHOT_SOURCE_DB_SHA256=<pre-launch-sha256> \
TTLAB_SCREENSHOT_SOURCE_DB_FAMILY_SHA256=<pre-launch-family-sha256> \
TTLAB_SCREENSHOT_RUNTIME_DB_PATH=/tmp/ttlab-capture-XXXX/papers.db \
TTLAB_SCREENSHOT_DATABASE_SNAPSHOT_EVIDENCE=/tmp/ttlab-capture-XXXX/database
    -snapshot.json \
TTLAB_SCREENSHOT_RUNTIME_INDEX_DIR=/tmp/ttlab-capture-XXXX/data/indexes \
node thesis/scripts/capture_interface_screenshots.mjs
make thesis

```

The runtime probe copies the SQLite database to a temporary directory, uses transient generation, verifies public routes, and confirms that anonymous admin access is denied. It never adds evidence rows to the source database. The screenshot command requires a clean committed source, pre-launch source hash, validated SQLite backup evidence, distinct runtime database inode, and exact copied index inventory. Its governance set stages the fail-closed public Finder and a protected aggregate Admin shell through a write-blocked service-reviewer session; the full set additionally requires an identity-current remediation-v2 dashboard panel. Both write only to a new external temporary staging directory, record capture/output/source-integrity assertions, and require separate privacy/rights-boundary and print-scale inspection before explicit promotion. The manuscript generator fails when a required frozen artefact is missing.

B.3 Evaluation commands

```

PYTHONPATH=backend .venv/bin/python -m app.evaluation.retrieval_experiment
PYTHONPATH=backend .venv/bin/python data/evaluation/
    validate_qa_faithfulness_v1.py
PYTHONPATH=backend .venv/bin/python data/evaluation/
    validate_recommendation_proxy_v1.py
PYTHONPATH=backend .venv/bin/python data/evaluation/
    validate_topic_author_silver_v1.py
PYTHONPATH=backend .venv/bin/python data/evaluation/
    validate_section_quality_silver_v1.py \
--database data/papers.db --evaluate-current

```

```
# Prospective suite: prepare freezes inputs before any final execution.
PYTHONPATH=backend .venv/bin/python \
  data/evaluation/run_peer_review_remediation_v2.py prepare
PYTHONPATH=backend .venv/bin/python \
  data/evaluation/run_peer_review_remediation_v2.py evaluate
PYTHONPATH=backend .venv/bin/python \
  data/evaluation/validate_peer_review_remediation_v2.py
```

Exact CLI arguments and phase-specific seeds are preserved in their manifests and the reproduction script. The prospective evaluator refuses changed frozen inputs, writes rights-sensitive raw outputs only to an ignored operator directory, and emits a closed allowlist of versionable structural outputs. The validator recomputes case identities, metrics, bootstrap intervals, and manuscript macros instead of trusting reported aggregates.

B.4 Application and document verification

```
PYTHONPATH=backend .venv/bin/python -m pytest
cd frontend && npm ci && npm test && npm run build && npm run test:e2e
cd ..
make thesis
.venv/bin/python scripts/validate_manuscripts.py --source-only
.venv/bin/python scripts/validate_manuscripts.py
pdffinfo build/thesis.pdf
pdffonts build/thesis.pdf
pdftoppm -png build/thesis.pdf tmp/pdfs/thesis
```

Local preflight verifies A4 thesis page size, metadata, embedded fonts, absence of Type 3 fonts and visible placeholder markers, citation/reference resolution, and serious overflow. PDF Express/Checker is external because it requires venue credentials.

B.5 Key identifiers

Table B.1 provides compact identifiers for checking that a rerun uses the intended corpus, indexes, model, and labels.

B.6 Availability boundary

The sanitized bundle contains code, tests, schemas, redistributable metadata, evaluation cases/labels, versionable structural results, prompts/configurations, code-native figure sources,

Table B.1: Key frozen identifiers. Full values are retained in manifests.

Artefact	Identifier
Historical corpus snapshot	corpus-04a010207327069a; full hash in the index manifests
Prospective technical snapshot	not run; hash prefix not run
Prospective protocol/datasets	not run / not run
Prospective package validation	manifest not available; attestation not available
Dense model	all-MiniLM-L6-v2 revision 826711e54e001c...a1def9
Dense model artifact	SHA-256 18309e334c02...048c0a7c
Retrieval labels	SHA-256 ccd16f6428c9...949a3cbe1

manuscript sources, hashes, and instructions. It excludes every PDF, SQLite database, run-time index, extracted/chunked publication text, rights-sensitive raw answer/recommendation/evidence file, unrestricted interface screenshot, raw full-text prompt, raw external XML, secret, and absolute local path. Exact full-corpus results are reproducible only for an authorized recipient who lawfully reacquires or receives the source material.

Schema and API Summary

C.1 Principal persisted contracts

The persisted contracts in Table C.1 define where provenance and review state survive beyond one request.

Table C.1: Principal persisted entities and traceability fields.

Entity	Principal fields
Paper	identifier; title; source authors; canonical identity links; metadata/provenance; ingestion, extraction, eligibility, OCR and review states; extraction/chunk/public-index generations; publication, rights and access states; timestamps
Chunk	identifier; paper foreign key; page start/end; inferred section; exact text; word/token count; source hash; extraction generation; eligibility/index status; timestamp
Index generation/manifest	immutable generation plus current pointer; corpus snapshot/hash; ordered chunk/source hashes; provider/model/revision; dimension/normalisation; configuration/code/index hashes; role/completeness
RAGAnswer	question; answer; retrieved chunk/paper/citation structures; provider/model/configuration; grounding and unsupported-claim warnings; reviewer fields; timestamps
Thesis Recommendation	transient request profile; ranked recommendation structures; evidence/gap/suggestion boundaries; citations; provider/model; grounding/review state; timestamp

Entity	Principal fields
Paper Artifact	paper foreign key; artifact type; generated text/JSON; citations and source sections; provider/model; grounding/review state
Canonical Author / Alias	canonical name; alias/provenance; identity/merge and review states; publication relation
Topic / PaperTopic / AuthorTopic	controlled name; paper/author links; score; source/evidence JSON; review state
ReviewEvent	item type/ID; previous/new state; reviewer ID/name/type/role; request ID; notes; field diff; previous/event hashes; timestamp

C.2 Principal API surfaces

Table C.2 separates public/transient functions from protected history and mutation operations.

Table C.2: User-facing API groups and security boundary.

Group	Purpose	Access
papers	approved statistics, browse/detail, extraction and artifact views	public projection
search	projection-filtered retrieval; technical scope exists only in protected/local code	public projection
ask	transient public generation; technical evaluation/persisted history separated	public projection; protected/local technical
recommendations	transient evidence-only/full output over approved public evidence	public projection
topics/authors	approved explorer relations; technical rebuild separated	public projection; protected rebuild
artifacts	public reviewed-safe output; generation/review operations	mixed
evaluation	evaluation state, corpus/index freshness, and public-safe results	public/protected
admin	review queues, mutations, events, corrections	public read
health/readiness	liveness and database/index completeness state	reviewer/admin bearer

FastAPI's generated OpenAPI document is available only in local-demo mode. Production disables interactive docs unless an operator creates an explicit boundary. Public request

models expose no technical-scope selector. Public serializers redact local paths, private reviewer notes, draft corrections, restricted source text, and technical identity hashes where appropriate.

C.3 Review-state semantics

`needs_review` denotes unreviewed content; `ai_reviewed` denotes declared AI formative review; `needs_reprocess` denotes preserved content that cannot be safely reproduced under current configuration; `reviewed`, `approved`, and `rejected` require a human actor with an appropriate role. Structural grounding remains separate from every review state.

C.4 Representative executable contracts

The following abridged listings show the implementation points behind the Finder, review, and privacy claims. They preserve operative field names and assertions while omitting imports, defaults not relevant to the claim, and test setup. The repository paths named in each explanation are authoritative; these excerpts are explanatory, not a second implementation.

Listing C.1: Abridged Finder request schema.

```
class ExtensionFinderRequest(BaseModel):
    interests: str = Field(min_length=1, max_length=2_000)
    skills: list[str] = Field(default_factory=list, max_length=50)
    available_time: Literal["2 weeks", "1 month", "semester"]
    project_type: Literal["software prototype", "data analysis",
                          "ML experiment", "dashboard/visualization", "other"]
    data_constraints: str = Field(max_length=1_000)
    preferred_difficulty: Literal["easy", "medium", "hard"]
    top_k: int = Field(default=5, ge=1, le=10)
    retrieval_mode: Literal["keyword", "feature_hashing",
                           "dense", "hybrid"]

    @field_validator("interests")
    @classmethod
    def interests_must_not_be_blank(cls, value: str) -> str:
        if not value.strip():
            raise ValueError("interests must not be empty")
        return value.strip()
```

Contract. Path: `backend/app/intelligence/extension_recommender.py`. **Rationale:** bound the public profile and make supported choices explicit. **Input/output:** untrusted profile JSON becomes a normalized typed request. **Failure:** blank, oversized, or out-of-enum val-

ues produce validation errors; declared skills, time, and data access remain unverified. **Trace:** RQ4 and UI-03.

Listing C.2: Abridged recommendation route boundary.

```
@router.post("/extensions")
def create_extension_recommendations(request, session):
    return recommend_extensions(session, request, persist=False)

@router.get("/extensions/history")
def extension_recommendation_history(
    session,
    _actor: Annotated[AuthenticatedActor, Depends(require_reviewer)],
):
    records = session.exec(select(ThesisRecommendation)).all()
    return [serialize_recommendation(record) for record in records]
```

Contract. **Path:** backend/app/api/recommendations.py; the prefix yields POST/api/recommendations/extensions. **Rationale:** separate public advice from review history. **Input/output:** a validated profile returns a transient recommendation; a reviewer-authenticated read returns serialized stored records. **Failure:** public history access is rejected and public generation cannot request persistence. **Trace:** RQ4/RQ7 and UI-03, UI-06, UI-09.

Listing C.3: Abridged review-event schema.

```
class ReviewEvent(SQLModel, table=True):
    review_event_id: str = Field(primary_key=True)
    item_type: str
    item_id: str
    action: str
    previous_status: str | None
    new_status: str | None
    reviewer_id: str
    reviewer_type: str
    reviewer_role: str
    request_id: str
    diff_json: dict[str, Any]
    previous_event_hash: str | None
    event_hash: str | None
    created_at: datetime
```

Contract. **Path:** backend/app/models/review.py. **Rationale:** retain actor attribution and correction history. **Input/output:** an item transition plus actor, request, diff, and prior hash becomes a canonical hash-linked event. **Failure:** database triggers reject updates/deletes; a chain mismatch is detectable, although the hash is neither a signature nor an external timestamp. **Trace:** RQ5/RQ7 and UI-07–UI-08.

Listing C.4: Core assertions from the public non-persistence test.

```
answer = client.post("/api/ask", json={"question": private_q})
finder = client.post("/api/recommendations/extensions",
                    json={"interests": private_interests})
assert answer.json()["persist"] is False
assert finder.json()["persist"] is False
assert answer.headers["cache-control"] == "no-store"
assert finder.headers["cache-control"] == "no-store"
assert client.get("/api/ask/history").status_code == 401
assert client.get("/api/recommendations/extensions/history").status_code == 401
assert session.exec(select(RAGAnswer)).all() == []
assert session.exec(select(ThesisRecommendation)).all() == []
```

Contract. **Path:** backend/tests/test_security_privacy.py. **Rationale:** falsify accidental retention or history disclosure. **Input/output:** private-looking public Ask/Finder requests return transient, non-cacheable responses. **Failure:** any persistence, cacheable header, or anonymous history response fails the test; external browser, proxy, operating-system, and network logs remain outside it. **Trace:** RQ7 and UI-06/UI-09.

Executed AI-Review Instruments

These AI-review instruments produced silver evidence, not human-participant data. Historical v1 records remain immutable. The prospective package freezes source-first cases/splits, repeats one AI process in shuffled order, retains disagreements, and versions only rights-safe structural outputs; raw answers, snippets, and recommendations remain restricted.

D.1 Retrieval relevance

For each question, the reviewer recorded category, answerability, relevant paper IDs, graded relevance, supporting chunk/page evidence, source rationale, creation decision, shuffled verification decision, disagreement, and adjudication. Rankings were unavailable to the label-creation process. Test cases were not used for tuning.

D.2 Answer and citation review

Every answer was segmented into atomic checkable claims. For each claim the instrument recorded support (supported/partial/unsupported), inline source IDs, correct paper/chunk/-page, citation relation (correct/partial/incorrect), unsupported general knowledge, and error tags. At answer level it recorded expected answer points (covered/partial/missing), answerability decision, abstention, returned-citation use, and notes.

D.3 Recommendation review

Each of three evidence-only and three full-finder results per profile was reviewed for paper relevance and source fidelity. Full outputs were also reviewed for fact/future-work/gap/suggestion separation, novelty caution, skills/time/data feasibility, MVP scope, stretch scope, risk, required skills, evaluation-plan quality, and usefulness as an AI proxy.

The proxy judgment is not evidence of student or supervisor benefit.

D.4 Topic and author review

Paper cases retained applicable controlled labels, explicit source passages, multi-label and other/unknown state, and two review passes. Author auditing first checked canonical/alias identity, excluded-paper leakage, authorship mismatch, and possible same-person candidates. Topic associations were then constrained to eligible indexed publications. No availability, endorsement, or supervision judgment was requested.

D.5 Generated-output review

Every stored RAG answer, recommendation, summary, technical summary, contribution, method, limitation, future-work section, suggested extension, skill/evaluation-plan structure, and podcast script was inspected for source locators, claim boundaries, configuration sufficiency, and safe status. Corrections were applied through append-only attributed events. AI review could assign only `ai_reviewed` or `needs_reprocess`; it could not approve content as human-reviewed.

D.6 Future human instrument boundary

A later human study would require an institutional ethics determination and a separate protocol covering recruitment, consent, inclusion criteria, sample-size rationale, assessor training, anonymity, retention, withdrawal, compensation, rating scales, agreement/adjudication, and adverse-event/correction handling. Nothing in the AI-silver process is presented as satisfying those requirements.

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